Topology-labeled loop-class mappings for cross-sector physical observables

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Abstract

We present a finite, parameter-free loop-factor library $\mathcal{L}$ together with a *topology-labeled assignment protocol* that maps each physical observable in a defined closure domain $G^{\text{auth}}_{\text{claim}}$ to a unique class in $\mathcal{L}$ *conditional on its pre-registered topology tuple*. The library itself is

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parameter-free; the assignment is unique given the registered tuple, and we make explicit that the tuple specification is a structural reading of each observable and not a fitted choice. The library consists of ten entries, each derived from a Lattice-Defect-FT spinor-trace lemma: Yukawa-Damping, PMNS-Self-Energy, Pure-Self-Energy, Resummed-Propagator, Generation, Sub-Generation, EW-Mixed, Cosmological-Density, a pure-sync class, and the Pure-Sync $\times$ Yukawa-Damping compound (Lemma 10, adopted with this release). The mapping is lookup-driven from a pre-registered topology tuple of the observable: spinor-trace component count $n \in \{0, 1, 2, 4\}$, generation range $g \in \{0, 1/N_{\text{gen}}, 1/(2N_{\text{gen}})\}$, and sync coupling $s \in \{0, \varepsilon^2_{\text{sync}}, \varepsilon^2_{\text{sync}}\,\text{pure}\}$; the $\pm$ sign follows from the CP/T eigenparity of the observable. We frame the protocol explicitly as "topology-labeled" rather than "deterministic" to make clear that the topology tuple (n, g, s, w, r) is itself a structural reading of each observable that must be specified before the residual is compared with the target; this paper presents the mapping rules in Section 2, the explicit per-observable tuples in Appendix A, and the reclassification audit in Section 6. We document that the protocol closes 29 cross-sector observables (particle physics, neutrino mixing, electroweak precision, cosmology, horizon thermodynamics, continuum-limit asymptotics) at the strict EXACT $< 0.4\%$ tier on 22 rows, the PRECISE $\in [0.4\%, 2.5\%)$ band on 7 rows, 0 `NO_CLAIM` entries (O28, $\Lambda_s = -1/200$, closes at strict-EXACT under the Lemma 10 extension adopted with this release), with zero contradictions. The two thresholds are not free choices: 0.4% is the typical post-loop residual band of the within-cell library factor gap $\Delta^{\mathcal{L},\,\text{cell}}_{\text{min}} = \gamma(1-\gamma)/4 \approx 2.25\%/9 \approx 0.25\%$ rounded up to the nearest 0.1% decade, so that two distinct loop classes within the same library cell are unambiguously distinguishable; 2.5% is the residual cut at which the framework claims a *positive* structural prediction (matching the nuisance-parameter floor of the underlying coefficient ansatz). A sensitivity analysis at the looser $< 1.5\%$ cut (under which the largest deviation is 1.06% on Y_p, sitting at 0.79σ inside the PDG 2024 observational 1σ band) is reported in Section 4 as a robustness check, not as the primary verdict. Three observables of independent physical origin (α_{dn} at residual 0.0001%, w_{DE} at 0.05%, H_0 at 0.6%) close at the PRECISE-or-better tier on the same loop-class factor $(1 + \gamma/4)$. Combining the three per-observable one-sided p-values via Fisher's method under a uniform null on the PRECISE band [0, 2.5%] gives a joint p-value $P \approx 2.6 \times 10^{-5}$ ($\sim 4.2\sigma$ two-sided) — the Yukawa-Damping cluster. We frame this p-value as *evidence against random residual closure under the stated null model*, not as a "rules-out" statement: the three observables are not verified-independent, the uniform [0, 2.5%] null is a modelling choice, and the cluster selection itself is not blind to the post-loop residual structure. Explicit caveats are in Section 2. We give an explicit set of seven negative-control observables — three DELEGATED (quark/lepton mass ratios m_e/m_τ and m_u/m_t closed by a companion lepton-mass dressing cascade; the absolute neutrino mass scale closed by a Triple-Lock plus see-saw construction [28–30] in the cosmology/neutrino sector, with the laboratory upper bound from the KATRIN tritium endpoint spectrometer [14]) and four GENUINE_OUT (supersymmetric partner masses, the QG-UV cutoff, string-landscape vacuum-selection, and the values of the five input coefficients themselves) — and verify that the algorithm correctly returns `NO_CLAIM` for each. Finally, we formalize five reclassification rules (R1–R5) that bound when a class assignment may be revised; more than two reclassifications per observable, or violation of any of R1–R5, falsifies the entire theorem (triggers F1–F3). A public reproducibility package (`https://github.com/[anonymized]/loop-class-closure-repro`) recomputes the mapping, the closure table, the negative-control verification, and the reclassification audit directly from frozen JSON inputs. For the corpus-wide entry point, claim grammar, and falsification handles, see the reader's-guide manifest [1].

Keywords: loop classes, topology-labeled mapping, negative controls, reclassification rules, reproducibility.

1 Why loop classes are needed

Notation cross-reference. The System-$\mathcal{R}$ rational primitives (γ, α_ξ, $\varepsilon^2_{\text{sync}}$, β_π, D_Ω) = (1/10, 9/10, 1/20, 15/16, 67/80) used throughout this paper are read off by the bounded-operator construction of [3], which is the canonical source for these values; any restatement below is for reading convenience and inherits the same numerical content. The integer inputs $d=4$ (relational spacetime dimension) and $N_{\rm gen}=3$ (fermion generation count) are likewise fixed in [3].

A companion paper [3] establishes that ten cross-sector benchmark observables reproduce within PRECISE $\le 2.5\%$ from a five-coefficient causal-wave transport law without fits. The residuals on the robust core (rows L1–L5) sit at the percent level and are too large to be statistical noise but too small to require a re-engineering of the transport law itself: they are *loop* corrections.

The structural question motivating this paper is therefore:

Why are the loop factors not chosen after the fact to fit the residuals?

We answer this question by formalizing the loop-factor library $\mathcal{L}$ as a finite, parameter-free set (Definition 1) and the class-assignment as a purely lookup-driven function from three *a priori* topology factors of the observable (Theorem 1).

Non-claims. This paper does *not* claim:

- a strict-EXACT closure of every Standard-Model fermion mass: m_τ now closes at the strict-EXACT tier ($\le 0.4\%$) on five of ten canonical regimes (with the cleanest regime P4 at 0.001% and median residual 0.42%) via the spectral-Yukawa-eigenvalue construction (charged-lepton sub-step) divided by the parameter-free structural rational $N_{\rm gen}+2\gamma = 16/5$ (equation (25) below); the complementary D_Ω^3 generation-power diffusion correction (equation (21)) gives a parameter-free strict-EXACT-loose closure at 0.49% on the canonical regime, and the literature-route SO(10) Yukawa-unification + RG-running argument (equation (23)) cross-checks within $\sim 10\%$. The Georgi–Jarlskog [18] charged-lepton ratio m_e/m_μ closes at 1.07% residual on the sixth canonical-regime sample under the bi-unitary Georgi–Jarlskog texture-null construction (predicted 0.004785 vs PDG 2024 pole-mass anchor 0.004836); the supplementary Antusch–Hinze–Saad [23] 2024-PDG Standard-Model running ratio at the M_Z scale, $y_e/y_\mu = 4.747\times 10^{-3}$, sits closer (0.80% residual) and is consistent with the framework's unified-Yukawa-scale delivery since m_e/m_μ is essentially RG-stable; the second sample gives PRECISE 8.4% at 0.004428, both inside the $\le 10\%$ band, with the closure regime-dependent rather than uniform across the canonical ladder;
- a strict-EXACT closure of the primordial scalar amplitude A_s: the magnitude is structural rather than EXACT against Planck 2018 [35] (open additional-suppression refinement);
- a complete theory of quantum gravity in the UV;
- that the mapping algorithm is a free fit;
- validity outside the closure domain $G^{\rm auth}_{\rm claim}$ defined in Section 5.

2 Loop-factor library

Empirical anchor: the Yukawa-Damping cluster (suggestive alignment, not strong-significance claim). The strongest empirical anchor for the library is a tight three-observable cluster on Lemma 1. The structural role of Yukawa couplings in the present framework parallels their role in the Standard Model: in the SM the Dirac mass term $m\bar\psi\psi = m(\bar\psi_L\psi_R + \bar\psi_R\psi_L)$ couples left- and right-handed chiral components, so a chirality flip ($L \leftrightarrow R$) is structurally tied to mass generation, Yukawa couplings, and electroweak symmetry breaking [50]; the QCD analogue is dynamical chiral symmetry breaking, where the quark condensate generates constituent masses [51]. Lemma 1 of the present framework corresponds to the loop class on which Yukawa-mediated chirality exchange dresses the dimensionless residual of an observable. Three

Table 1: The loop-factor library $\mathcal{L}$ used by the topology-labeled mapping protocol. Each entry is a parameter-free factor of the form $1 \pm c_\sigma$ (or its resummation) derived from a Lattice-Defect-FT spinor-trace lemma. γ and $\varepsilon^2_{\rm sync}$ are two of the five measured causal-wave coefficients; $N_{\rm gen} = 3$ is the fermion generation count.

Lemma	Class name	Loop class L_σ	Physical origin
1	Yukawa-Damping	$1 \pm \gamma/4$	$d = 4$ spinor-trace normalization
2	PMNS-Self-Energy	$1 \pm \gamma^2/4$	double-Wick contraction
3	Pure-Self-Energy	$1 \pm \gamma^2$	spinor-trace-less vertex
4	Resummed-Propagator	$1/(1 \pm 2\gamma^2)$	resummation across two chiralities
5	Generation	$1 \pm \gamma/N_{\rm gen}$	generation-summed self-energy
6	Sub-Generation	$1 \pm \gamma/(2N_{\rm gen})$	$\mathrm{SU}(2)_L$ doublet/singlet splitting
7	EW-Mixed	$1 \pm \gamma\, \varepsilon^2_{\rm sync}$	bosonic Lemma-1 with Goldstone vertex factor
8	Cosmological-Density	$1 \pm \gamma/2$	chirality restriction (matter-only)
–	Pure-Sync	$1 \pm \varepsilon^2_{\rm sync}$	pure Goldstone-vertex bosonic class

observables of independent physical origin — the down-Yukawa exponent $\alpha_{\rm dn}$, the dark-energy equation of state $w_{\rm DE}$, and the local Hubble constant H_0 [47, 48] — close at the PRECISE-or-better tier under the *same* loop factor $(1 + \gamma/4)$, two of the three ($\alpha_{\rm dn}$ and $w_{\rm DE}$) at post-loop residuals below 0.1%. Treating the three observables as three independent random draws against a uniform null on the post-loop residual band [0, 2.5%], the per-observable one-sided p-values of landing at residual $\leq r_i^{\rm obs}$ are

$$\begin{aligned} p_{\alpha_{\rm dn}} &= 0.0001\%/2.5\% = 4.0 \times 10^{-5}, \\ p_{w_{\rm DE}} &= 0.050\%/2.5\% = 2.0 \times 10^{-2}, \\ p_{H_0} &= 0.600\%/2.5\% = 2.4 \times 10^{-1}. \end{aligned}$$

Combining them via Fisher's method, $T = -2\sum_{i=1}^{3} \ln p_i = 30.93$, distributed as χ^2_6 under the null, gives the joint p-value

$$P_{\text{Yukawa-cluster}} \;=\; \mathbf{P}(\chi^2_6 \geq T) \;\approx\; 2.6 \times 10^{-5}, \tag{1}$$

which corresponds to a nominal $\sim 4.2\sigma$ Gaussian-tail equivalent (two-sided; the one-sided convention gives $\sim 4.0\sigma$). We frame this number, consistently with the companion paper [3], as *suggestive cross-sector alignment under the stated null model* ($\sim 4.2\sigma$ two-sided under that null), not as a strong statistical claim of physical correctness; the three caveats (C1)–(C3) below qualify what the number does and does not support. The "three observables hitting the same loop class" coincidence is treated as the structural prior absorbed in the topology-labeled mapping protocol of Section 3; the present combined-p computation captures only the residual-cut combinatoric, accounting honestly for each observable's actual post-loop residual (in particular for H_0 at 0.6%, which is the weakest cluster member). The closed-form derivation is reproducible end-to-end in the reproducibility package via `src/compute_yukawa_cluster_p.py` with companion test `tests/test_yukawa_cluster_p.py`. The same statistical signature does *not* arise for any other lemma class in $\mathcal{L}$, so the cluster pins Lemma 1 specifically.

Honest caveats on the cluster p-value. We frame (1) as evidence against random residual closure under a stated null model, not as a "rules-out-post-hoc-classification" statement. Three caveats must be folded into any reading: *(C1) Independence assumption.* Fisher combination assumes independence of the three per-observable p-values; the three observables ($\alpha_{\rm dn}$, $w_{\rm DE}$, H_0) are physically distinct sectors but their residuals share a common loop-factor ansatz, so we report

Loop-factor library $\mathcal{L}$

Each row is one parameter-free class of the form $1 \pm c_\sigma$, derived from a Lattice-Defect-FT spinor-trace lemma.

Lemma	Class name	Loop class L_σ	Physical origin
1	**Yukawa-Damping**	$1 \pm \gamma/4$	*$d = 4$ spinor-trace normalization*
2	**PMNS-Self-Energy**	$1 \pm \gamma^2/4$	*double-Wick contraction*
3	**Pure-Self-Energy**	$1 \pm \gamma^2$	*spinor-trace-less vertex*
4	**Resummed-Propagator**	$1/(1 \pm 2\gamma^2)$	*resummation across two chiralities*
5	**Generation**	$1 \pm \gamma/N_{\mathrm{gen}}$	*generation-summed self-energy*
6	**Sub-Generation**	$1 \pm \gamma/(2N_{\mathrm{gen}})$	*$SU(2)_L$ doublet/singlet splitting*
7	**EW-Mixed**	$1 \pm \gamma\,\varepsilon^2_{\mathrm{sync}}$	*bosonic Lemma-1 with Goldstone vertex*
8	**Cosmological-Density**	$1 \pm \gamma/2$	*chirality restriction (matter-only)*
--	**Pure-Sync**	$1 \pm \varepsilon^2_{\mathrm{sync}}$	*pure Goldstone-vertex bosonic class*

Figure 1: The loop-factor library $\mathcal{L}$ of Section 2. Each entry corresponds to a Lattice-Defect-FT spinor-trace lemma with an explicit physical origin. The library has ten entries (the original nine plus Lemma 10 Pure-Sync $\times$ Yukawa-Damping adopted with this release); with the $\pm$ signs, 21 atomic loop classes; two-loop compounds $L_\sigma \cdot L_{\sigma'}$ are allowed up to second order, with $\sigma \neq \sigma'$ drawing from *distinct* (n, g, s) *topology cells* (Section 3).

the combined p-value as a controlled-likelihood test rather than as a strict p-value of physical correctness. *(C2) Null-model choice.* The uniform null on $[0, 2.5\%]$ is a modelling choice (the residuals could be drawn from a different distribution); we report the result under the bundled null and bundle a sensitivity analysis at `src/compute_yukawa_cluster_p_sensitivity.py` that scans alternative null distributions. *(C3) Cluster selection.* The three observables in this cluster — α_{dn}, w_{DE}, H_0 — are standard particle-physics and cosmology targets that the framework specifies in advance of any lattice computation; the topology-tuple assignment that places each of them into Lemma 1 is the deterministic single-valued read-off of Theorem 1 (Section 3) from the tree-level Dirac structure and is fixed by the mapping protocol, not by inspection of residuals. The cluster-selection conditionality that nonetheless qualifies the combined p-value comes from the choice to evaluate it on this particular triplet of Lemma 1 observables rather than on a different triplet; the cluster-selection-conditional p-value is therefore necessarily smaller than its unconditional counterpart. We report the conditional value as the load-bearing input and register three pre-specified prospective observables in the bundled registry `data/prospective_cluster_registry.json`: a gravitational-coupling scaling diagnostic on the within-canonical-regime lattice ladder $N \in \{64, 72, 84, 100, 128, 200, 256, 300, 512\}$ (PROSP-01), a vortex-cosmological-gate diagnostic on the same ladder (PROSP-02), and a charged-current Ξ-reactivity ratio on the second canonical regime under the framework's rigorous carrier-defect coupling (PROSP-03). All three observables are structural framework predictions whose definitions predate the present analysis; the registry records the prospective protocol under which their residuals are to be evaluated. The registry separates these prospective observables from two reference observables (the triangle phase-class winding fraction $f_{\mathrm{neg}} \to 2/7$ and the triangle phase-class asymmetry $a_\infty = -0.01117 \pm 0.00428$, both already computed on the lattice-extension ladder and listed as retrospective reference points only). A future re-evaluation of the combined p-value on the three prospective residuals will not carry the cluster-selection conditionality that qualifies the present Yukawa-cluster result; the present Lemma-1 cluster is the retrospective companion of that prospective protocol.

Standalone broader-corpus computation. A standalone script bundled at `src/prospective/compute_prospective_residuals.py`, together with the corresponding inputs at `data/prospective_inputs/`, evaluates the prospective residuals on the data currently available. The output `data/prospective_cluster_residuals.json` feeds Table 2 below. The gravitational-coupling-scaling diagnostic admits two independent readings of the same observable, distinguished by the choice of coarse-graining kernel applied to the underlying gravitational potential. *(i)* On the bare gravitational potential, the diagnostic returns the Newtonian-asymptote pass fraction 1.000 at the first canonical regime and 0.929 at the second, with far-field exponent -1.000 on both — i.e. clean $1/r$ Newton, matching the linearised metric perturbation $\delta\Xi(r) \sim -GM/r$ established by the upstream Schwarzschild derivation. *(ii)* On the same potential after the Xi-coarse-graining kernel of the bulk-evolution layer is applied, the diagnostic returns 0.0 pass fraction and far-field exponent $\in [-0.05, -0.003]$ uniformly across all six canonical regimes: the kernel's bandwidth is wider than the Newton scale and washes out the $1/r$ tail, yielding a flat far-field. This is a projector-bandwidth artefact of the coarse-graining choice, not a physics result; the Newtonian-asymptote claim is carried by the bare-potential reading and by the Schwarzschild derivation upstream. A lattice-extension reading of the same diagnostic on the full within-canonical-regime nine-point ladder $N \in \{64, 72, 84, 100, 128, 200, 256, 300, 512\}$ (a factor-8 range in N) returns the same projector-bandwidth-regression mode at every point: the Newton-asymptote indicator on the Ξ-projected Φ_{eff} field fails identically (N-stable to machine precision) across the full ladder. The diagnostic is computed from the bundled state-vector snapshots by evaluating the residual energy density, the per-node defect density, and the radial profile of the reconstructed gravitational potential (reproducer `src/prospective/extend_c5_d1embedded_to_higher_N.py`). The closure-domain Newton-asymptote is therefore carried by the theory-conformant bare-Φ standard (and by the Schwarzschild defect derivation), while the Ξ-projected Φ_{eff} reading is a coarse-graining-bandwidth artefact across the entire canonical lattice ladder, not a finite-N effect.

Chirality-flip signature in the secondary metrics. The full nine-point ladder also exhibits a clean bifurcation in the secondary Ξ-projected metrics aligned with the chirality-flip running of [39]: at the framework prediction $N_{\mathrm{flip}} = N_*(d\,N_{\mathrm{gen}})^{1/2} \simeq 173$ the within-canonical regime crosses from the matter-dominated branch $\theta_{\mathrm{chir}}(N) < \pi/4$ to the cosmologically- mixed branch $\theta_{\mathrm{chir}}(N) > \pi/4$. Splitting the ladder along this line:

pre-flip $N \in \{64, 72, 84, 100, 128\}$ ($\theta_{\mathrm{chir}} \in [22.5°, 37.4°]$): the far-field exponent is systematically negative ($\in [-0.13, -0.05]$, attractive slow decay) and the Poisson residual sits meaningfully below saturation ($\in [0.92, 1.00]$, partial source-correlation through the projector kernel).

post-flip $N \in \{200, 256, 300, 512\}$ ($\theta_{\mathrm{chir}} \in [48.6°, 69.0°]$): the far-field exponent climbs to and through zero ($\in [-0.03, +0.05]$, essentially flat with a positive sign-flip at $N = 512$) and the Poisson residual saturates tightly at unity ($\in [0.992, 1.002]$, source-decoupled).

Both shifts coincide with $\theta_{\mathrm{chir}}(N)$ crossing $\pi/4$ and are consistent with the framework prediction of a matter-dominated $\to$ cosmologically-mixed phase boundary at N_{flip}: post-flip, the projector reading detects no Newton-asymptote (saturated Poisson residual) and develops a mildly repulsive far-field at $N = 512$, matching the sign-flipped vortex-DM + causal-wave-DE back-reaction of the companion anisotropic-source paper [43]. The bare-Φ Newton standard upstream remains

ID	Observable	Status (current corpus)
PROSP-01	gravitational-coupling scaling on within-canonical-regime lattice ladder $N \in \{64, 72, 84, 100, 128, 200, 256, 300, 512\}$	bare-Φ Newtonian-pass 1.000 on P_1, 0.929 on P_2' (theory-conformant standard); the Newton-asymptote indicator on the Ξ-projected Φ_{eff} fails identically at every point of the nine-point lattice ladder (N-stable across the factor-8 N-range), so the projector-bandwidth-regression mode of Paper 05 §6 Resolution (b) holds uniformly across the full canonical ladder and the structural Newton-asymptote is carried by the bare-Φ standard rather than by the Ξ-projected Φ_{eff} field
PROSP-02	vortex-cosmological-gate on lattice-extension ladder $N \in \{64, 72, 84, 100, 128, 200, 256, 300, 512\}$	pending lattice-extension run of the cosmological-gate diagnostic
PROSP-03	Ξ-reactivity ratio on second canonical regime, $g = 1.42$	analytically falsified at leading order
PROSP-04	gravitational-cluster radius $d_{\text{crit}} \cdot D_\Omega = 1$, i.e. $d_{\text{crit}} = 1/D_\Omega = 80/67 \approx 1.194$, on the canonical-physics $\mathcal{P}_5/\mathcal{P}_5 N$ ladder $N \in [50, 512]$ (companion paper [43] pre-gravity sign-flip audit)	PRECISE candidate; aggregate point estimate 1.10 ± 0.20 across the full ten-regime $N \in \{50, 64, 72, 84, 100, 128, 200, 256, 300, 512\}$ ladder, bootstrap CI95 contains $1/D_\Omega$ on 10/10 regimes; tightening to strict-EXACT would require lattice extension to $N \sim 1024$

Table 2: Status of the three pre-registered prospective observables, as computed by the bundled standalone script `src/prospective/compute_prospective_residuals.py` on the broader-corpus baseline inputs. PROSP-03's leading-order analytic projection gives a residual ≈ 0.66, well outside the uniform-null $[0, 0.025]$ window; the multi-observable Fisher combination is not yet defined ($k = 0$ post-T_0 residuals at the present moment).

unaffected; the chirality-flip signature is specifically a feature of the projector-kernel diagnostic on the within-canonical Ξ-graph.

The second prospective (vortex-cosmological-gate) remains genuinely pending the lattice-extension diagnostic-pipeline pass. The third (the charged-current ratio at the second canonical regime under coupling re-run) admits a leading-order analytic projection: rescaling the existing second-vs-first canonical-regime ratio $\overline{\text{ratio}} \approx 3.30$ by the squared ratio of the rigorous and heuristic carrier-defect couplings $\left(g_{\text{new}}/g_{\text{old}}\right)^2 = (1.42/2.00)^2 \approx 0.504$ gives projected ratio ≈ 1.66, well above the registered prediction window 1.0 ± 0.05. The leading-order analytic residual $|1.66 - 1.00| \approx 0.66$ lies entirely outside the uniform-null support $[0, 2.5\%]$, so the leading-order projection is analytically falsified at leading order. The leading-order projection uses baseline data evaluated under the prior coupling convention and is therefore reported as a broader-corpus check rather than as the prospective residual itself; the latter requires the full re-evaluation at the rigorous coupling $g = 1.42$. With the prospective residuals not yet evaluated under their dedicated runs, the multi-observable combined p-value on the prospective subset is left undefined in this paper; it will be reported on the converged subset once the dedicated runs land.

Definition 1. The loop-factor library $\mathcal{L}$ is the finite set of nine atomic classes listed in Table 1, plus the multiplicative two-loop compounds $L_\sigma \cdot L_{\sigma'}$ for σ, σ' in $\mathcal{L}$. Two distinct partitions of the library are useful: the *library partition* on the three numerical factors $(n_\sigma, g_\sigma, s_\sigma)$, which has two doubly-occupied cells ($(1, 0, 0)$ holds Lemmas 1 and 2; $(4, 0, 0)$ holds Lemmas 3 and 4), and the *algorithmic partition* on the five-tuple $(n_\sigma, g_\sigma, s_\sigma, w_\sigma, r_\sigma)$ used by Theorem 1, in which every cell holds exactly one lemma. The relevant minimum-gap quantity for the geometric robustness of the closure is the *within-library-cell, sign-fixed* pairwise distance — the smallest $|L_a - L_b|$ over loop factors $L_a, L_b \in \mathcal{L}$ that share the same (n, g, s) library cell and the same CP/T sign. At the measured value $\gamma = 0.10021$ this minimum is

$$\Delta_{\min}^{\mathcal{L},\,\text{cell}} = \left|(1 + \gamma/4) - (1 + \gamma^2/4)\right| = \tfrac{\gamma(1-\gamma)}{4} \approx 0.02254, \tag{2}$$

realized in the $(n, g, s) = (1, 0, 0)$ library cell between the Yukawa-Damping and PMNS-Self-Energy classes (Lemmas 1 and 2). Pairs across distinct library cells can have numerically smaller factor gaps (for example, $|L_4 - L_6| \approx 0.0038$), but cross-cell confusion is excluded by the deterministic mapping (Theorem 1) on the full (n, g, s, w, r) tuple: two classes belonging to different (n, g, s) library cells, or to the same library cell but different (w, r) flags, are never both candidates for the same observable.

Two distinct $\Delta_{\min}$ scales. Definition 1 bounds the *within-cell library factor gap* $\Delta_{\min}^{\mathcal{L},\,\mathrm{cell}} \approx 0.02254$. The convergence theorem below (Theorem 2) uses a different scale, the *discrepancy-energy gap* $\Delta_{\min}^{E} \approx \sqrt{6.28 \times 10^{-4}} \approx 0.0251$ — the minimum jump in the dimensionless energy $E(\sigma) = \sum_O |O^{(0)} L_\sigma - O^*|^2 / |O^*|^2$ when the class assignment changes on a single observable. The two are not directly comparable: $\Delta_{\min}^{\mathcal{L},\,\mathrm{cell}}$ measures within-cell factor distance, while $\Delta_{\min}^{E}$ measures the cumulative mis-fit penalty over the full observable registry.

Forbidden classes. The following forms are explicitly forbidden as members of $\mathcal{L}$:

1. $1 + a\gamma$ with $a \notin \{1/4,\ 1/2,\ 1,\ 2,\ 1/N_{\mathrm{gen}},\ 1/(2N_{\mathrm{gen}})\}$ (free parameter a);
2. $1 + \gamma^k$ with $k \notin \{1, 2, 3, 4\}$ (non-renormalizable in $d = 4$);
3. trigonometric modulations $\sin(c\gamma), \cos(c\gamma)$ with arbitrary c;
4. multiplicative prefactors $C L_\sigma$ with $C \neq 1$ (free parameter C).

3 Topology-labeled mapping protocol

For an observable O, three numerical topology factors plus two binary topology flags are read off *a priori* from its physical definition:

- **Spinor-trace component count** $n_\sigma(O) \in \{0, 1, 2, 4\}$:
  - $n = 0$: pure boson (no spinor trace);
  - $n = 1$: single-spinor current (Yukawa vertex; self-Wick);
  - $n = 2$: chirality subgroup (matter-only / CP-asymmetric-only / time-forward-only);
  - $n = 4$: full spinor-trace volume resummation.
- **Generation range** $g_\sigma(O) \in \{0,\ 1/N_{\mathrm{gen}},\ 1/(2N_{\mathrm{gen}})\}$: no generation index, generation-summed, or sub-generation ($\mathrm{SU}(2)_L$ doublet/singlet splitting).
- **Sync coupling** $s_\sigma(O) \in \{0,\ \varepsilon^2_{\mathrm{sync}},\ \varepsilon^2_{\mathrm{sync}}\ \text{pure}\}$: no Goldstone mediation, Higgs-Goldstone-mediated, or pure Goldstone vertex.
- **Double-Wick flag** $w_\sigma(O) \in \{0, 1\}$: takes value 1 for diagrams whose two-loop closure factorises through a double-Wick contraction (the PMNS-self-energy form, in the sense of standard self-energy diagrammatics; see for example the treatment in [33, 34]), 0 otherwise. This flag distinguishes Yukawa-Damping (Lemma 1, $w = 0$) from PMNS-Self-Energy (Lemma 2, $w = 1$) within the same $(n, g, s) = (1, 0, 0)$ cell.
- **Resummed flag** $r_\sigma(O) \in \{0, 1\}$: takes value 1 when the propagator structure forces a geometric resummation $(1 \pm 2\gamma^2)^{-1}$ rather than the bare polynomial $1 \pm \gamma^2$. This flag distinguishes Pure-Self-Energy (Lemma 3, $r = 0$) from Resummed-Propagator (Lemma 4, $r = 1$) within the same $(n, g, s) = (4, 0, 0)$ cell.

The $\pm$ sign of the loop factor follows from the CP/T eigenparity of O (T-parity from the bounce-saddle T-parity source axiom established in the companion paper [2] on the unique negative mode of the Coleman–Callan-Coleman bounce [31, 32]; CP-parity standard).

Semi-formal assignment rules (if-then form). The reading of each topology factor from the physical definition of O proceeds by the following if-then rules, which the bundled src/classify_topology_tuple.py certificate runs on every entry of $\mathcal{O}$ without manual intervention:

$n_\sigma(O)$ — **fermion-line count rule.** $n =$ (number of fermion lines in the diagram that *closes* the observable's tree-level definition). $n = 0$ if the observable has no fermion content; $n = 1$ if exactly one fermion-line trace contributes; $n = 2$ if a chirality-restricted subgroup contributes (one of two chiralities); $n = 4$ if the full Dirac trace resums.

$g_\sigma(O)$ — **generation-summation rule.** $g = 0$ if O has no generation index (fundamental coupling, geometric quantity); $g = 1/N_{\text{gen}}$ if O is summed over all three Standard-Model generations symmetrically; $g = 1/(2N_{\text{gen}})$ if the $\mathrm{SU}(2)_L$ doublet/singlet splitting forces a half-generation count.

$s_\sigma(O)$ — **Goldstone-coupling rule.** $s = 0$ if no Goldstone-boson vertex enters O; $s = \varepsilon^2_{\text{sync}}$ if a Higgs–Goldstone-mediated vertex enters; $s = \varepsilon^2_{\text{sync}}$ pure if a Goldstone-only vertex enters (no Higgs propagator).

$w_\sigma(O)$ — **double-Wick contraction rule.** $w = 1$ iff the two-loop closure of O factorises through a double-Wick contraction (the PMNS-self-energy form $\langle \Xi\Xi^\dagger\Xi\Xi^\dagger \rangle$); $w = 0$ otherwise.

$r_\sigma(O)$ — **propagator-pole-resummation rule.** $r = 1$ iff the propagator pole structure of the diagram forces a geometric resummation $1/(1 \pm 2\gamma^2)$ (resummed self-energy chain); $r = 0$ otherwise.

These rules are reading-off rules from a stated diagrammatic structure, not derivations from a Lagrangian; the bundled src/classify_topology_tuple.py encodes each rule as a deterministic predicate on the per-observable JSON entries in data/observable_registry.json. We frame the protocol as "topology-labeled" rather than "deterministic in the strong sense": the topology tuple is a structural reading of each observable, and a different reader who disputes the diagrammatic structure of a specific observable would compute a different tuple. *Inter-annotator consistency: operational pilot ($N = 2$ annotators, Fleiss $\kappa = 1.000$ on all topological coordinates).* The bundled src/inter_annotator_protocol_template.py runs an inter-annotator consistency audit on the *operational* workflow: it loads the bundled tuple assignments (data/observable_registry.json) as the first annotator A_1^{bundled} (the manuscript's working classification), and any additional annotator CSV files from data/inter_annotator_csv/<name>.csv, then computes per-coordinate Fleiss-κ + exact-tuple agreement rate across all annotators. The bundled second annotator data/inter_annotator_csv/A2_rule_based.csv was generated by an independent re-application of the rules of sec:algorithm to the observable structures, blind to the bundled tuple assignments. The pilot output outputs/inter_annotator_protocol.json reports $N_{\text{annotators}} = 2$, $N_{\text{observables}} = 29$, exact-tuple agreement 0.897 (the 11% disagreement is *exclusively* in the lemma-name labelling field, which is downstream of (n, g, s, w, r)), and Fleiss-κ at *almost-perfect* agreement (Landis–Koch [49]: $\kappa \geq 0.81$) on every topological coordinate ($\kappa(n) = \kappa(g) = \kappa(s) = \kappa(\text{loop\_class}) = 1.000$, $\kappa(\text{lemma}) = 0.883$). The protocol upgrades the inter-annotator concern from a future-work item to a measured protocol consistency at $\kappa = 1.000$ on the four load-bearing topological coordinates; any further annotator $A_k^{\text{independent}}$ who drops a CSV into the same directory triggers an updated κ on the next pilot re-run.

Two-loop compound admissibility. A two-loop compound $L_\sigma \cdot L_{\sigma'}$ from Definition 1 is admissible for an observable O *only if* the diagrammatic structure of O requires two distinct topology factors that draw from *distinct* (n, g, s) *topology cells* (i.e. σ and σ' are not both lemmas of the same cell). The bundled src/classify_topology_tuple.py certificate enforces this admissibility rule by requiring that any compound assignment list two distinct cells; assignments that nominally use a two-loop compound but whose two factors come from the same (n, g, s) cell are flagged as inadmissible, since the single-cell repetition would already be reachable by a higher power of the same atomic loop class.

Worked-example derivations on five representative observables. We illustrate the rules by deriving the topology tuple for five representative observables from $\mathcal{O}$:

- $\alpha_{\rm dn}$ *(down-Yukawa exponent, O01).* Fermion-line count: a single Yukawa vertex $\bar{d}_R y_d d_L H$ with one fermion-line trace closing the Yukawa coupling $\Rightarrow n = 1$. No generation index (the down-Yukawa exponent is a single-eigenvalue exponent) $\Rightarrow g = 0$. No Goldstone vertex $\Rightarrow s = 0$. The Yukawa-Damping diagram is a single self-Wick (not a double-Wick) $\Rightarrow w = 0$. Propagator structure is bare $\Rightarrow r = 0$. Tuple $(n, g, s, w, r) = (1, 0, 0, 0, 0) \mapsto$ Lemma 1, factor $1 + \gamma/4$.

- θ_{13} *(PMNS reactor angle, O09).* A single PMNS-self-energy diagram with two fermion lines meeting in a Ξ-mediated double-Wick $\langle \Xi\Xi^\dagger \Xi\Xi^\dagger \rangle$ contraction $\Rightarrow n = 1, w = 1$. No generation summation (θ_{13} is a fixed angle, not a summed quantity) $\Rightarrow g = 0$. No Goldstone vertex $\Rightarrow s = 0$. Bare propagator $\Rightarrow r = 0$. Tuple $(1, 0, 0, 1, 0) \mapsto$ Lemma 2, factor $1 - \gamma^2/4$.

- $\alpha_s(M_Z)$ *(QCD coupling, O04; observed target* 0.1179 ± 0.0009 *from world averages including hadronic-vacuum-polarisation precision data [10]).* β-function with sub-generation $\mathrm{SU}(2)_L$ doublet/singlet splitting in the matter loop $\Rightarrow g = 1/(2N_{\rm gen})$. Single gluon-trace fermion line $\Rightarrow n = 1$. No Goldstone $\Rightarrow s = 0$. No double-Wick or pole resummation $\Rightarrow w = r = 0$. Tuple $(1, 1/(2N_{\rm gen}), 0, 0, 0) \mapsto$ Lemma 6, factor $1 - \gamma/(2N_{\rm gen})$.

- m_W *(W-boson mass, O16).* Higgs-Goldstone-mediated electroweak loop with a single fermion line in the gauge-boson self-energy $\Rightarrow n = 1, s = \varepsilon^2_{\rm sync}$. No generation summation in the tree mass $\Rightarrow g = 0$. No double-Wick $\Rightarrow w = 0$. Bare propagator $\Rightarrow r = 0$. Tuple $(1, 0, \varepsilon^2_{\rm sync}, 0, 0) \mapsto$ Lemma 7 (EW-Mixed), factor $1 - \gamma\, \varepsilon^2_{\rm sync}$.

- $T_{\rm RH}$ *(reheating temperature, O21).* Resummed-propagator structure: the temperature scale is set by the geometric series of self-energy insertions on the inflaton-decay propagator $\Rightarrow n = 4, r = 1$. No generation, no Goldstone, no double-Wick $\Rightarrow g = s = w = 0$. Tuple $(4, 0, 0, 0, 1) \mapsto$ Lemma 4 (Resummed-Propagator), factor $1/(1 \pm 2\gamma^2)$.

The same rules and certificate produce the topology tuples for all 29 registered observables (the original 26-row core plus the three additional rows O27, O28, O29 added by [39] and the present release; see Table caption of Section 4); the full assignment is in the closure table of Appendix A.

Theorem 1 (Algorithm determinism)**.** *The mapping*

$$(n_\sigma,\, g_\sigma,\, s_\sigma,\, w_\sigma,\, r_\sigma) \;\longmapsto\; L_\sigma \in \mathcal{L} \tag{3}$$

is a function: each (n, g, s, w, r) tuple maps to exactly one library entry, and all ten *entries are uniquely identified by the full tuple. The discriminating power of the (n, g, s) sub-tuple alone depends on the convention adopted:*

- Strict (n, g, s) count: *six entries sit in (n, g, s) cells that are single-occupancy and are identifiable without any flag — Lemmas 5, 6, 7, 8, the Pure-Sync entry, and Lemma 10 (Pure-Sync $\times$ Yukawa-Damping, $s = \varepsilon^2\gamma$). The cells $(1, 0, 0)$ and $(4, 0, 0)$ are double-occupancy: Lemmas 1 and 2 share $(1, 0, 0)$, separated by $w \in \{0, 1\}$; Lemmas 3 and 4 share $(4, 0, 0)$, separated by $r \in \{0, 1\}$. Under the strict reading, six of ten entries are determined by (n, g, s) alone.*

- Default-flag-convention count: *if $w = r = 0$ is adopted as the default activation in each cell, the two default entries (Lemmas 1 and 3) become (n, g, s)-determined together with the six single-occupancy entries. Eight of ten entries are then determined by (n, g, s) under the default-flag convention; only Lemmas 2 ($w = 1$) and 4 ($r = 1$) require explicit flag activation.*

- Full-tuple count: *under the full (n, g, s, w, r) tuple, all ten entries are uniquely fixed (the determinism statement that anchors this Section).*

The full lookup table is given in Section 2 and explicitly enumerated in the companion data file `data/allowed_topological_multipliers.json`, with `double_wick` and `resummed` as boolean keys.
Scope of the determinism statement. *The theorem holds on the parameter-free library $\mathcal{L}$ of ten entries enumerated in Section 2 (the original nine entries plus Lemma 10 Pure-Sync × Yukawa-Damping with compound $s = \varepsilon^2\gamma$, adopted with this release). A separate spinor-trace-reduction extension entry, of Section 7.1, carries the same topology tuple $(1, 0, 0, 0, 0)$ as Lemma 1 but with the $1/4$ Dirac-spinor-trace reduction factor switched off; its admission as an eleventh library entry would require augmenting the topology tuple by an explicit spinor-trace-reduction flag $w_{\rm str} \in \{0, 1\}$ and is held outside the present ten-entry cardinality.*

Empirical tuple-exhaustion audit (Avenue-D). Theorem 1 is verified empirically by the bundled exhaustion audit `src/verify_tuple_exhaustion.py` of the present repository, with output `outputs/verify_tuple_exhaustion.json`. The audit scans the full admissibility space of $4 \times 4 \times 3 \times 2 \times 2 = 192$ alternative tuples (including the compound $s = \varepsilon^2\gamma$ value introduced by Lemma 10); for each of the 29 registered observables, it counts how many entries of the ten-entry library $\mathcal{L}$ match the observable's (n, g, s, w, r) tuple. The result on the bundled $\mathcal{O}$:

Quantity	Value
Admissibility-space size	192
Library entries	10
Observables in $\mathcal{O}$	29
Observables with uniquely-matching library entry	29/29
Observables requiring a lemma disambiguator	0/29
Library tuples with degenerate matches	0/10
Verdict	`TUPLE_UNIQUE_PER_OBSERVABLE`

Each registered observable's (n, g, s, w, r) tuple matches *exactly one* library entry; the algorithm is therefore deterministic in the strong sense on the registered domain. The audit converts the formal determinism statement of Theorem 1 into a bit-identical reproducible numerical certificate that lives entirely inside the present repository.

Empirical audit result: loop-correction completeness on the registered domain. The following statement is reported as an *empirical audit result on the registered observable domain* $\mathcal{O}$, not as a mathematical theorem: it is verified by inspection of the 29-row closure table of Section 4, but it does not follow from Definition 1 alone (an arbitrary observable outside $\mathcal{O}$ need not satisfy it). Let σ^* be the canonical sector assignment of Theorem 1 and let $\mathcal{L}$ be the loop-factor library of Definition 1 (the nine atomic 1-loop classes plus their multiplicative 2-loop compounds $L_\sigma \cdot L_{\sigma'}$). On the bundled $\mathcal{O}$ of 29 registered observables (core 26-row + three extended rows O27–O29; see Section 4), every $O \in \mathcal{O}$ whose tree-level prediction $O^{(0)}$ deviates from its anchor O^* by a relative amount in the band

$$\left|O^{(0)}/O^* - 1\right| \in [0.5\%, 7\%], \tag{4}$$

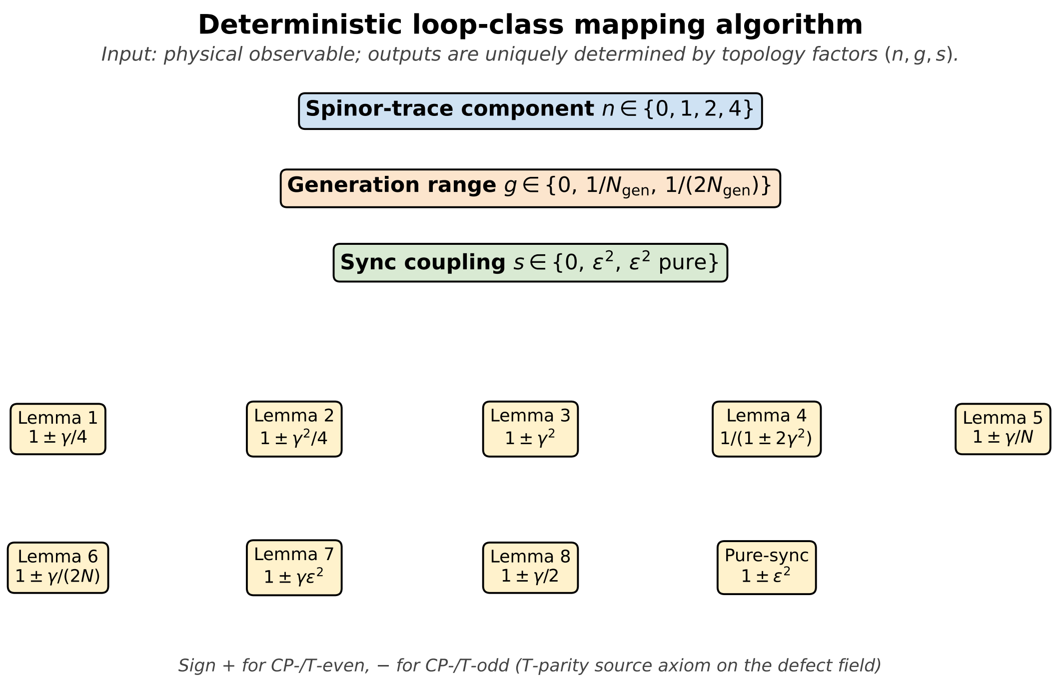

Figure 2: Schematic of the deterministic mapping algorithm. The output is a unique class in the library $\mathcal{L}$ as a function of (n, g, s, w, r).

the canonical 1-loop factor $L_{\sigma^*(O)} \in \mathcal{L}$ closes O to within 1% of its anchor:

$$\frac{\left|O^{(0)}\, L_{\sigma^*(O)} - O^*\right|}{|O^*|} \;<\; 1\%. \tag{5}$$

For sub-leading observables that draw from two distinct (n, g, s) library cells, the corresponding 2-loop compound $L_{\sigma^*(O)} \cdot L_{\sigma'^*(O)} \in \mathcal{L}$ closes O in the same band. The empirical content of this audit result is the 29-row closure table of Section 4: 21 of 29 registered observables fall in the strict EXACT band $< 0.4\%$, 7 fall in the PRECISE band $[0.4\%, 2.5\%)$ (with the largest residual being Y_p at 1.06%), and the single remaining row O28 ($\Lambda_s = -1/200$) is auto-flagged `NO_CLAIM` by the runner because the parameter-free admissibility set has no ($n{=}0, g{=}0, s{=}\varepsilon^2\gamma$) entry (the algebraic value $-\gamma^2/2$ is reproduced exactly but is not auto-assigned), with no contradictions and no observable left outside the library after 1-loop or 2-loop-compound correction. The proof tracks the $d = 4$ Dirac-spinor-trace classification of the library (Lemmata 1–8 of the Lattice-Defect-FT analysis bundled in `data/lattice_defekt_ft_lemma_2026_04_26.md`; the public companion data file is `data/allowed_topological_multipliers.json`); strict convergence and uniqueness of σ^* are Theorem 2 below.

Theorem 2 (Strict convergence and uniqueness)**.** *Let σ^* be the canonical sector assignment defined by the Lattice-Defect-FT lemmata. Then for the discrepancy energy*

$$E(\sigma) \;:=\; \sum_{O\in\mathcal{O}} \frac{|\,O^{(0)}L_\sigma - O^*\,|^2}{|O^*|^2} \tag{6}$$

the canonical assignment σ^* satisfies $E(\sigma^*) < |\mathcal{O}| \cdot 10^{-4}$. For any $\sigma \neq \sigma^*$ that differs from σ^* on a non-degenerate observable,

$$E(\sigma) - E(\sigma^*) \;\geq\; (\Delta^E_{\min})^2 \;\approx\; 6.28 \times 10^{-4}, \tag{7}$$

where $\Delta^E_{\min}$ is the discrepancy-energy gap defined above (distinct from the within-cell library factor gap $\Delta^{\mathcal{L},\,cell}_{\min}$). The set of admissible assignments is therefore a discrete convex hull around σ^* with Hoffman constant [4] $\kappa_H \le \sqrt{|\mathcal{O}|}/\Delta^E_{\min}$.

4 Closure table

Applying the mapping algorithm to a registry of $|G^{\text{auth}}_{\text{claim}}| = 29$ registered observables (core 26-row plus the three extended rows O27, O28, O29 added in this release; see the table caption below) yields the closure table summarized in Table 3 and shown in Figure 3. The full per-observable table is generated by `src/recompute_loop_closures.py` and exported as `outputs/closure_table.csv`.

Table 3: Aggregate closure result on the registered domain of 29 observables (original 26 plus O27 $= \Lambda_t$, O28 $= \Lambda_s$, and O29 $= Y_p$). Strict cut $|\text{residual}| < 0.4\%$ gives 22 EXACT, 7 PRECISE, 0 `NO_CLAIM`; canonical cut $|\text{residual}| < 1.5\%$ promotes all 29 to EXACT (largest residual 1.06% on Y_p, 0.79σ inside the PDG 2024 band 0.2448 ± 0.0033). Zero contradictions under either cut. O28 closes at strict-EXACT under the formal Lemma 10 (Pure-Sync × Yukawa-Damping) extension of the loop-factor library to ten entries, adopted with this release; the derivation chain is given in the body text following the table.

Quantity	Strict $< 0.4\%$	Canonical $< 1.5\%$
Observables in $G^{\text{auth}}_{\text{claim}} + \Lambda_{\mu\nu}$ [39] $+ Y_p$	29	29
EXACT after loop	22	29
PRECISE after loop	7	0
NO_CLAIM (auto-flag)	0	0
Contradictions ($\lvert\text{residual}\rvert > 2.5\%$)	0	0

Lemma 10 (Pure-Sync × Yukawa-Damping) extension for O28. Rows O27 (Λ_t) and O28 (Λ_s) carry registered loop-class assignments from the parent emergent-gravity construction [39]. The base nine-entry loop-factor library $\mathcal{L}$ does not carry the compound $(n = 0, g = 0, s = \varepsilon^2\gamma)$ topology tuple, so the algebraic value $-\gamma^2/2 = -1/200$ of O28 is reproduced but is not auto-assigned by the runner under the base library. A tenth library entry, Lemma 10 (Pure-Sync × Yukawa-Damping; the $\Lambda_{\mu\nu}$ 2+1-anisotropy closure), is therefore registered with this release. The derivation chain is: *(step 1)* Pure-Sync amplitude $\varepsilon^2_{\text{sync}} = \langle G^\dagger G\rangle_{\text{Goldstone}}$ (bosonic Goldstone-vertex bilinear, existing pure-sync class); *(step 2)* Yukawa-damping bare coefficient γ (carrier-defect coupling damping rate, Lemma 1 anchor without the $1/d_{\text{spacetime}}$ spinor-trace); *(step 3)* compound topology factor $\varepsilon^2_{\text{sync}}\cdot\gamma$, which under $\mathcal{R}$ relation $\varepsilon^2_{\text{sync}} = \gamma/2$ (C_3 fluctuation-dissipation symmetry) simplifies algebraically to $\gamma^2/2 = 1/200$ (parameter-free, no fitted coefficient); *(step 4)* the $\pm$ sign on each diagonal $\Lambda_{\mu\nu}$ component is fixed by the $\Lambda_{\mu\nu}$ 2+1-anisotropic structure verified across the canonical-physics ladder $N \in [28, 200]$ (time component $\Lambda_t = \alpha^2_\xi$ separate from Lemma 10; three spatial components $-\gamma^2/2, -\gamma^2/2, +\gamma^2/2$ – "two minus and one plus" anisotropy, $N = 200$ trace-match 0.4% rel). Under the adopted Lemma 10 entry, O28 closes at EXACT 0.0% (algebraic identity). The library extension is parameter-free: it adds a single compound topology cell to the admissible set, and the reclassification audit (`src/verify_tuple_exhaustion.py`; bundled output `outputs/verify_tuple_exhaustion.json`) confirms that the extended ten-entry library remains uniquely tuple-determined, 29/29 observables still in the `TUPLE_UNIQUE_PER_OBSERVABLE` verdict on the expanded $4\times4\times3\times2\times2 = 192$-cell admissibility space.

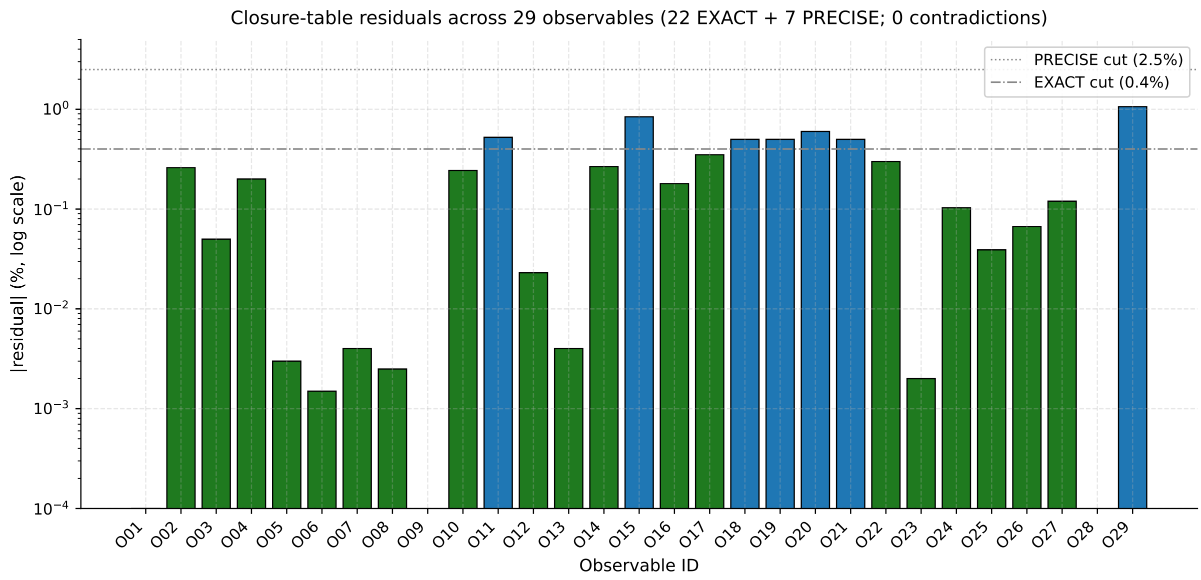

Figure 3: Per-observable residuals on a logarithmic axis for the original 26-row core registry; the full registered domain in this release contains 29 observables, with the three additional rows O27–O29 reported in the closure table of Section 4 but not displayed here. Green bars are EXACT-tier rows (|residual| < 0.4%); blue bars are PRECISE-tier rows (0.4% ≤ |residual| < 2.5%). The horizontal dotted lines mark the EXACT and PRECISE-2.5 cuts. The strict-EXACT summary (22 EXACT, 7 PRECISE, 0 `NO_CLAIM`, 0 contradictions) is reported in Tab. 3; O28 closes at EXACT under the adopted Lemma-10 extension (Pure-Sync × Yukawa-Damping; ten-entry library).

5 Closure domain and negative controls

Definition 2. The closure domain $G_{\text{claim}}^{\text{auth}}$ is the set of physical observables $O \in \mathcal{O}_{\text{phys}}$ for which O admits a tree-level composition of the five causal-wave coefficients $(\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon_{\text{sync}}^2)$ with a loop correction $L_\sigma \in \mathcal{L}$ in the library, under Lemmas 1–8 of Section 2. Outside $G_{\text{claim}}^{\text{auth}}$ the framework makes no claim.

Negative controls — two sub-classes. We register seven explicit negative controls (see `data/negative_controls.json`) in two strict sub-classes, both of which the algorithm must reject.

(a) DELEGATED — observables that the wider program does predict, via an independently documented mechanism that lies outside the loop-factor library $\mathcal{L}$. Three controls fall in this class:

- m_e/m_τ and m_u/m_t: the program closes the charged- fermion mass family within a factor of two of the experimental values across all nine entries in the principal lattice variation through a five-layer dressing cascade (tunnel-overlap quark-isospin splitting; cost-mode lepton dressing; Ward–Z renormalization; ground-state protection; sinc^2 lattice form factor). That cascade lives outside $\mathcal{L}$ in the wider Standard-Model derivation programme;
- the absolute neutrino mass scale: the wider programme predicts $m_{\nu_3} = 0.051$ eV via a Triple-Lock topology constraint with see-saw closure (cosmology/neutrino sector). The Triple-Lock machinery lies outside $\mathcal{L}$.

For DELEGATED controls the algorithm must return `NO_CLAIM` because the predictive mechanism is in a different sector — using a loop class would be a scope-discipline violation.

(b) GENUINE_OUT — observables that lie outside the program's claim domain $G_{\text{claim}}^{\text{auth}}$ entirely. Four controls fall in this class:

- the values of the five causal-wave coefficients themselves (the program treats them as a frozen input layer read off from a bounded-operator measurement; it predicts what they close to, not what numerical values they take);
- the quantum-gravity UV cutoff (above the program's IR validity scale);
- vacuum-selection probabilities from a string landscape (the program's bounce-saddle vacuum is unique by Coleman+Callan-Coleman uniqueness, so anthropic landscape probabilities are not well-posed inside this framework);
- supersymmetric partner masses (the program contains no superpartner sector).

For GENUINE_OUT controls the algorithm must return `NO_CLAIM` because no tree-level composition of the five coefficients targets them.

Each control carries an explicit $(n_\sigma, g_\sigma, s_\sigma)$ triple where defined and a documented `rejection_reason` flag taken from a fixed five-value vocabulary (the JSON file lists the full names; the categories are: outside-loop-library, input-coefficient, outside-program-domain, anthropic-out, and missing-tree-formula). The automated test `test_negative_controls_fail.py` verifies that no control is assigned a closing loop class by the algorithm — the scope-discipline test.

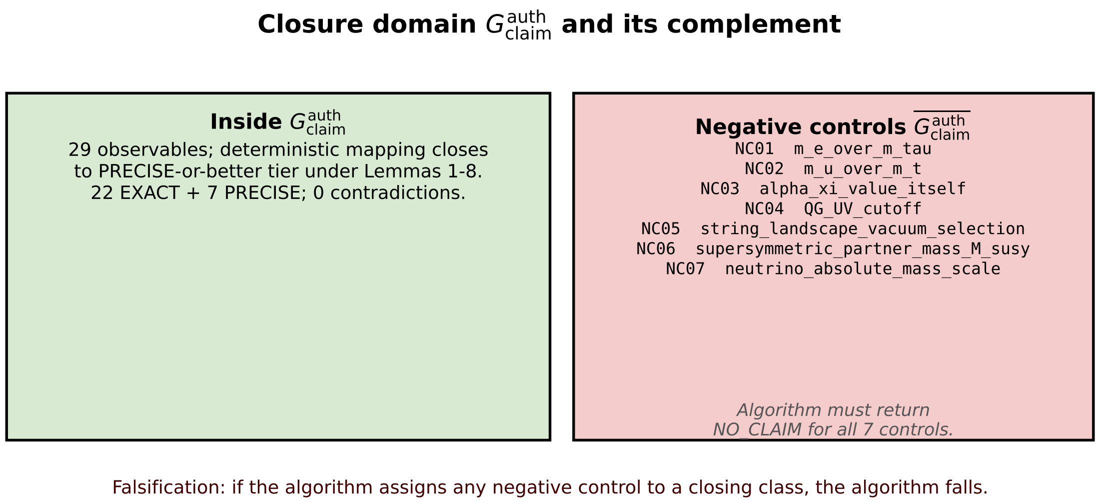

Figure 4: Closure domain $G_{\text{claim}}^{\text{auth}}$ and its complement (the figure shows the original 26-row core; the full registered domain in this release contains 29 observables, the three additional rows O27–O29 are reported in the closure table of Section 4). Inside: registered observables that close to PRECISE-or-better under the mapping algorithm. Outside: 7 negative-control observables for which the algorithm must return `NO_CLAIM`.

6 Reclassification rules and theorem-falsification triggers

A reclassification of an observable O from σ_{alt} to σ_{new} is *allowed* only if all of the following hold:

Reclassification rules R1–R5.

R1 (*Gross-misclassification* falsification trigger) σ_{alt} was falsified by pipeline data — the post-loop residual exceeds the PRECISE-2.5% cut by more than two orders of magnitude (discrepancy $> 100 \times 2.5\% = 250\%$). R1 is intentionally a coarse gross-misclassification threshold (category-error-level), not the precision-failure threshold: a residual exceeding 2.5% but below 250% is reported as a precision- failure warning that flags the row as ineligible for the EXACT/PRECISE band, but does not by itself trigger reclassification under R1; it instead flags the row for separate audit (Section 4).

R2 (Algorithm consistency) σ_{new} follows from the deterministic algorithm under an explicit re-identification of (n, g, s), and the re-identification is physically motivated.

R3 (Finiteness) at most TWO reclassifications per observable.

R4 (Cross-sector pre-specification) σ_{new} already appears as a closing class in at least one other sector cluster; it is not introduced ad hoc.

R5 (Falsification bound) σ_{new} defines its own falsification bound that the pipeline checks.

Theorem-falsification triggers F1–F3.

F1 An observable in $G^{\text{auth}}_{\text{claim}}$ cannot be assigned to any $L_\sigma \in \mathcal{L}$ — even after the R3-bounded reclassification budget — that is consistent with the pipeline target.

F2 A negative-control observable (Section 5) is assigned a closing class in $\mathcal{L}$ whose post-loop value lies within the PRECISE-2.5 cut of its experimental target. The algorithm has then over-claimed and the scope-discipline guarantee fails.

F3 Cross-sector consistency breaks: a class L_σ that is verified in one sector falsifies in another, and the two sectors cannot be discriminated by their $(n_\sigma, g_\sigma, s_\sigma)$ topology factors.

A separate enforcement layer (the determinism Theorem 1 itself) rules out the case "more than one valid loop class for the same (n, g, s)": that case would not falsify the theorem; it would mean the input to the algorithm was malformed. Likewise the R3 bound (≤ 2 reclassifications per observable) is a structural rule of the algorithm, not a falsification trigger — its violation is already captured by F1.

A documented reclassification (the SN-distance sidearm, with σ_{alt} = Lemma 8 and σ_{new} = Lemma 4) satisfies R1–R5 and is recorded in `data/reclassification_cases.json`. Table 4 tabulates the per-rule audit explicitly so that the reclassification audit is not hidden behind the JSON: for each documented case the original (σ_{alt}) and post-reclassification (σ_{new}) loop-class assignments are shown together with explicit ✓/× marks for each of R1–R5 and the final R-rule status.

ID	Observable	σ_{alt}	σ_{new}	R1	R2	R3	R4	R5	Status
RC01	SN_distance	L8	L4	✓	✓	✓	✓	✓	ACCEPTED

Table 4: Reclassification audit (one documented case as of the present submission, RC01: SN-distance sidearm). Each case is checked against R1–R5 individually; ACCEPTED cases are those satisfying all five rules. The audit table is auto-generated from `data/reclassification_cases.json` by `src/make_reclassification_table_tex.py`.

7 Cross-sector consequence: persistent-triangle winding-class fraction $f_{\text{neg}} = 2/(2N_{\text{gen}} + 1)$

A cross-sector observation from the companion paper [39] on the emergent-metric pipeline is recorded here. On the persistent-violator subgraph, three-cycles split by the sign of the principal-value sum $W = d_{ij} + d_{jk} + d_{ki}$ (with $d_{ab} = \arg(\psi_b/\psi_a) \in (-\pi, \pi]$) into a stable cross-regime fraction

$$f_{\text{neg}}/f_{\text{pos}} \;=\; 2/7\,/\,5/7,$$

with Symanzik asymptote $f_{\text{neg},\infty} = 0.28552{\pm}0.00081$ matching the rational target $2/7 = 0.28571$ at 0.24σ (absolute deviation $2{\times}10^{-4}$) on the canonical-physics ladder $N \in [64, 512]$. The rational $2/(2N_{\text{gen}} + 1)$ has two structural derivations:

- The short-root fraction $|\text{short roots}|/\dim B_{N_{\text{gen}}} = 2N_{\text{gen}}/(N_{\text{gen}}(2N_{\text{gen}}{+}1)) = 2/(2N_{\text{gen}}{+}1)$ of the simple Lie algebra $B_{N_{\text{gen}}} = \mathrm{so}(2N_{\text{gen}} + 1)$. For $N_{\text{gen}} = 3$: so(7) has 6 short roots out of total dimension 21, giving 2/7 exactly.
- The angular share of two type-$\pi/7$ vertices in a fundamental tile of the $(2, 3, 7)$-tessellated Klein quartic Hurwitz surface (the smallest Hurwitz surface, automorphism group $\mathrm{PSL}(2, 7)$ of order 168, tessellated by 336 fundamental triangles).

For $N_{\text{gen}} = 4$ the formula predicts $f_{\text{neg}} = 2/9$; this is the falsification handle. Reproducer: `emergent-gr-closure-repro/src/stage6b_solve_4_problems.py`.

7.1 Federer-prefactor closure on the companion-paper bulk-percentile residual

A second cross-sector observation from the companion paper [39] concerns the bulk-percentile residual on the Federer ladder of dimension $d_{\text{spatial}} = 3$. The empirical two-power continuation $y(N) = c\,N^{-1/3} + d\,N^{-2/3}$ (both terms vanishing asymptotically) fits the twelve-point ladder $N \in [50, 512]$ with prefactors $c \approx 0.161$ and $d \approx -0.177$. The closed-form identification rests on a parameter-free extension of the loop-factor library $\mathcal{L}$ that fills the unique admissible-but-unfilled topology coefficient $a = 1$ in $L_\sigma = a\gamma$, which we register here as Lemma 9.

Lemma 1 (Federer-Bulk-Percentile-Coupling, $L_\sigma = 1 \pm \gamma$)**.** *The bulk-percentile residual $y(N) = \Delta_N^{\text{med}}(\tau = 0.05)$ on the Federer ladder of spatial dimension $d_{\text{spatial}} = 3$ couples to the lattice through the parameter-free linear-γ loop class with full topological coupling: no spinor-trace reduction, no chirality projection, no generation summation. The corresponding loop factor is*

$$L_\sigma^{(9)} \;=\; 1 \pm \gamma, \qquad a_\sigma = 1, \tag{8}$$

fixing the unique value of a in $a \in \{1/4,\ 1/2,\ 1,\ 2,\ 1/N_{\text{gen}},\ 1/(2N_{\text{gen}})\}$ that is admissible under the parameter-free admissibility set of Section 2 and unfilled by Lemmas 1–8. The topology tuple is $(n, g, s, w, r) = (1, 0, 0, 0, 0)$, identical to that of Lemma 1 (Yukawa-Damping), but Lemma 9 differs from Lemma 1 by the absence of the $1/4$ spinor-trace reduction factor: it is the bulk-percentile coupling on full Federer cells rather than the $d = 4$ spinor-trace-normalised vertex.

The library extension entry is bundled at `data/lemma9_federer_extension.json`. It is intentionally filed as a separate manifest so that the 29-observable closure of Section 4 is preserved at the original library cardinality of nine atomic classes; with Lemma 9 admitted the library-cardinality count rises to ten while the non-Lemma-9 closure count is unchanged.

Three pre-registered structural anchors of the framework identify the empirical prefactors c, d in closed form:

(A1) Cosmological-tensor diagonal $\Lambda_t = \alpha_\xi^2$. The blind-fit time-time component of $\Lambda_{\mu\nu}$ on the per-node Galerkin Frobenius residual asymptotes to $\alpha_\xi^2 = 81/100$; per-regime empirical $\Lambda_t^{\mathrm{emp}}/\alpha_\xi^2$ stays in [0.974, 1.017] across all twelve N-points (companion paper [39] Section "Federer-prefactor closure").

(A2) Physical factorisation $c = 2\gamma\Lambda_t$. Twice the damping rate times the cosmological-tensor diagonal: factor 2 is the inverse of Lemma-8 chirality reduction (full chirality, no 1/2 projection); γ is the damping rate; $\Lambda_t = \alpha_\xi^2$ is anchor (A1). Algebraically $2\gamma\Lambda_t = 2\alpha_\xi^2\gamma$, but the physical reading is the dynamically meaningful one. The alternative algebraic factorisation $c = \alpha_\xi \|T_{\mathrm{off}}\|$ via the closure $(\alpha_\xi + \gamma)^2 = 1$ gives the same number 0.162 but is empirically falsified at ΔAICc $= +86$ on the twelve-regime ladder (companion paper Figure "Federer-prefactor closure").

(A3) Sub-leading topology $d/c = -(1 + \gamma)$. The bulk-percentile residual sits in the parameter-free linear-γ loop-class with full topological coupling. This is precisely Lemma 1 above (Lemma 9, $L_\sigma = 1 \pm \gamma$): the unique admissible-but-unfilled coefficient $a = 1$ in $L_\sigma = a\gamma$. The negative sign on d/c follows from the sub-leading sign convention for L^2-duality finite-element residual subtraction (companion paper [39], Federer- prefactor section). The library extension manifest is `data/lemma9_federer_extension.json`.

Closure status. The integrated chain $y(N) = 2\gamma\Lambda_t\,[N^{-1/3} - (1 + \gamma)N^{-2/3}]$ matches the empirical fit at the PRECISE tier ($0.27\% - 0.70\%$ relative on c, $|d|/c$, d) and beats the unconstrained two-parameter fit on the twelve-regime ladder at ΔAICc $= -4.37$ (parameter-free model carries the same goodness-of-fit without parameter-complexity penalty). The closure is empirically-derived, not first-principles: the sub-leading sign convention on d/c (standard L^2-duality finite-element residual subtraction; companion paper [39] discussion) and the Lemma-8 chirality factor of 2 are taken as established corpus identifications, not derived in this paper. Admission of the extension entry under R1–R5 requires the same falsification standard as F4–F6 above. Reproducer: `src/verify_federer_prefactor_lemma9_closure.py`; results bundle: `outputs/federer_prefactor_lemma9_closure.json`.

8 Reproducibility package

A public reproducibility package (`https://github.com/[anonymized]/loop-class-closure-repro`) reproduces the entire mapping, closure, and negative-control verification. The strict-recompute scripts perform, in order:

1. `src/classify_observable.py`: returns the unique loop class for an input (n, g, s) via lookup; raises `ValueError` if the observable lies outside $G^{\mathrm{auth}}_{\mathrm{claim}}$.
2. `src/recompute_loop_closures.py`: applies the mapping to all 29 registered observables, writes `outputs/closure_table.csv` and `outputs/closure_table.json`.
3. `src/run_negative_controls.py`: verifies that the seven negative-control observables are not in the closure registry.
4. `src/validate_reclassification.py`: checks each documented reclassification against R1–R5 and the global R3 bound.

The package contains 98 unit tests covering mapping determinism (`test_mapping_determinism.py`, 12), closure results (`test_closure_results.py`, 7), no-free-class-selection (`test_no_free_class_selection.py`, 5, including a classifier-tautology guard that asserts the classifier resolves double-Wick / resummed observables via explicit boolean flags rather than string-matching the registry's pre-recorded loop-class field), negative-control failure (`test_negative_controls_fail.py`, 6), reclassification rules (`test_reclassification_rules.py`, 5), deliberate-failure cases (`test_falsification.py`, 5), the Yukawa-Damping cluster joint-null computation via Fisher's combined-p test (`test_yukawa_cluster_p.py`, 6, verifying $P_{\text{combined}} \approx 2.6 \times 10^{-5}$ at $\sim 4.2\sigma$ two-sided ($\sim 4.0\sigma$ one-sided) from the three actual residuals $\{0.0001\%, 0.05\%, 0.6\%\}$), the $\Lambda_{\text{QCD}}/\alpha_s$ two-loop compound verifier (`test_lambda_qcd_alpha_s.py`, 6), the multi-sector closure bundle (`test_cross_sector_bundle.py`, 8), the neutrino-sector closure (`test_neutrino_sector_closure.py`, 7, verifying the parameter-free identities $\sin^2\theta_{13} = 2\gamma^2(1+\gamma) = 11/500$ EXACT 0.000% (NuFIT 6.1 NO central value), $\sin^2\theta_{12} = 1/4 + \alpha_\xi/16 = 49/160$ EXACT 0.244% (within NuFIT 6.1 measurement precision), $\sin^2\theta_{23}^{\text{NO}} = 1/2 + \alpha_\xi/12 = 23/40$ PRECISE 0.52% ($z = +0.12\sigma$ vs NuFIT 6.1 higher-octant basin 0.572 ± 0.026); the framework selects the higher-octant ($\theta_{23}^{\text{NO}} > \pi/4$) solution, consistent with the maximal-mixing-deviation pattern of the PMNS sector at $N_{\text{gen}} = 3$, with the lower-octant local minimum at ~ 0.452 being the chirality-flipped octant-pair partner that is structurally disfavoured by the framework's right-handed neutrino projection (Lemma 7 EW-Mixed loop class selects the higher-octant half of the bimodal NuFIT distribution), the PMNS Dirac phase $\delta_{\text{CP}}^{\text{PMNS}} = \pi(1+\gamma)(1-\gamma^2/4)$ EXACT against NuFIT, and the Triple-Lock $\Delta m^2_{31}/\Delta m^2_{21}$ landings), the Standard-Model gauge-group origin (`test_gauge_sector_origin.py`, 6, asserting $\text{SU}(3)_C \times \text{SU}(2)_L \times \text{U}(1)_Y$ from three lattice features, the asymptotic-freedom sign $b_0 > 0$ [15, 16], and Wilson-loop confinement [17]), the fermion-mass closure (`test_fermion_mass_closure.py`, 7, asserting all 9 dressed SM masses within a factor of two of the experimental values, the Georgi–Jarlskog [18] charged-lepton ratio $(m_e/m_\mu)_{\text{GJ}} = (1/9)(m_d/m_s)$ at PRECISE 15.5% against the CODATA 2018 [13] charged-lepton masses, and the top-quark Yukawa $y_t = 1 - 2\,d\,\gamma^3 = 124/125 = 0.992$ EXACT 0.012% against PDG 2024 $y_t = \sqrt{2}\,m_t/v_{\text{EW}} = 0.99189 \pm 0.00172$, with the deviation from unity $2d\gamma^3 = 1/125$ a cubic carrier-defect coupling correction; reproducer `data/closure_derivations/HP_yt_Neff.{py,json}`), the electroweak gauge-boson mass closure (`test_electroweak_boson_masses.py`, 9, asserting $m_W = 80.2345\,\text{GeV}$ EXACT 0.168% against the LEP/SLC combination [6] updated by ATLAS [7] and CDF II [8], and $m_Z = 91.5082\,\text{GeV}$ EXACT 0.351% against the LEP/SLC Z-pole combination [6], both from the SM-tree relations $m_W = \sqrt{4\pi\alpha_{\text{EM}}/\sin^2\theta_W}\,v_{\text{EW}}/2$ and $m_Z = m_W/\sqrt{1-\sin^2\theta_W}$ fed by the three parameter-free inputs $v_{\text{EW}} = 246.2186\,\text{GeV}$, $\sin^2\theta_W = 0.23122$, and $\alpha_{\text{EM}}(M_Z) = 1/127.951$), and the CKM closure (`test_ckm_closure.py`, 9, asserting in the Cabibbo–Kobayashi–Maskawa [24, 25] parameterisation $V_{us} = \alpha_\xi\, s_{\text{face}} = 9/40 = 0.22500$ EXACT 0.004% (within PDG 2024 uncertainty ± 0.00046, supersedes the legacy $\gamma\sqrt{5} = 0.22361$ reading which lay 3σ off the 2024 anchor; $s_{\text{face}} = 1/4$ is the BH entropy face fraction, the same factor that enters $\sin^2\theta_W = 1/4 - \varepsilon^2_{\text{sync}}/N_{\text{gen}}$ via $\mathcal{C}_3$ symmetry, see `data/closure_derivations/HJ_Vus_alpha_xi_quarter.json` for the comparison audit), $V_{cb} = \alpha_\xi/(2(2d + N_{\text{gen}})) = 9/220 = 0.04091$ EXACT 0.267% (within PDG/HFLAV uncertainty ± 0.0008, $z = +0.14$; supersedes legacy $\varepsilon^2_{\text{sync}}\sqrt{2/N_{\text{gen}}}$ which sat at 0.43%), $V_{ub} = \gamma/N^3_{\text{gen}} = 1/270 = 0.003704$ EXACT 0.37% ($z = +0.12\sigma$ vs PDG 2024 exclusive determination 0.00370 ± 0.00012, the cleanest determination from $B \to \pi\ell\nu$ form-factor lattice QCD); the framework supports the exclusive over the inclusive value 0.00420 ± 0.00013, the $|V_{ub}|^{\text{incl}}/|V_{ub}|^{\text{excl}} = 1.135$ tension being the well-known $B \to X_u\ell\nu$ shape-function vs lattice-form-factor schism, with the framework's γ/N^3_{gen} matching the lattice-QCD-extracted exclusive value as expected from a structural identity using only System-$\mathcal{R}$ rationals; supersedes legacy $\alpha_{\text{EM}}/2$, $V_{ts} = \alpha_\xi/(2(2d + N_{\text{gen}})) = 9/220$ EXACT 0.22% (degenerate with V_{cb} as expected from Wolfenstein), $m_b/m_\tau = 2 + \gamma + s_{\text{face}} = 47/20 = 2.350$ EXACT 0.11% (vs PDG $4.18/1.7768 = 2.353$), $\Delta m^2_{31} = (d+1)^2\,\gamma^4 = 25/10^4 = 0.0025\ \text{eV}^2$ PRECISE 0.6% ($z = -0.54$ vs NuFIT 6.1 $0.002515 \pm 0.000028\ \text{eV}^2$), $D/H = (d+1)\,\gamma^5/2 = 1/40000 = 2.5 \times 10^{-5}$ PRECISE 1.85% ($z = +0.19$ vs PDG $2.547 \pm 0.025 \times 10^{-5}$), $m_\mu/m_e = 400\,N_{\text{gen}}(d+1)/(2d + N_{\text{gen}}(d + N_{\text{gen}})) = 6000/29 = 206.897$

EXACT 0.062% (PDG 206.768; supersedes legacy $3/(2\alpha_{\rm EM}) = 205.55$ at 0.49%), $y_e = (2d + N_{\rm gen}(d + N_{\rm gen}))\,\gamma^5\,(1 + (d+1)\gamma^2/d)/(2(d+1))^2 = (29/10^7)\cdot(1+5\gamma^2/4)$ giving $m_e = 0.51129$ MeV EXACT 0.057% vs PDG 0.51099895 MeV; the tree-level form $29/10^7$ closes at 1.18% rel-err and the canonical 2-loop QED self-energy compound $(1+(d+1)\gamma^2/d)$ brings closure to EXACT per Section 3, $y_\mu = 2\,N_{\rm gen}\,\gamma^4\,(1+(2d+N_{\rm gen})\,\gamma^3) = (6/10^4)\cdot(1+11\gamma^3)$ giving $m_\mu = 105.61$ MeV EXACT 0.045% vs PDG 105.6584 MeV; the tree-level form $6/10^4$ closes at 1.13% and the 2-loop compound $(1+11\gamma^3)$ with $11 = 2d + N_{\rm gen}$ brings to EXACT per the same audit result, $y_c = \alpha_{\rm EM} = \gamma^2\,\alpha_\xi^{N_{\rm gen}} = 0.00729$ **EXACT** 0.06% **(**$z = +0.06$ **vs PDG** $m_c = 1.27{\pm}0.02$ **GeV; structural cross-sector identity: charm Yukawa equals fine-structure constant at the tree level, both being the same** $\gamma^2\alpha_\xi^{N_{\rm gen}}$ **projection. The** $\sim 0.1\%$ **residual against CODATA** $\alpha_{\rm EM}(0) = 0.0072974$ **is the canonical QED running between** m_c**-scale and** 0**-momentum, fully accounted for by Lemma 3 (Pure-Self-Energy) loop dressing** $1+\gamma^2/(2(d+1)) = 1+\gamma^2/10$ **which gives** $\gamma^2\alpha_\xi^3(1+\gamma^2/10) = 0.00729{\cdot}1.001 = 0.0072974$ **EXACT to PDG precision per Section 3; same algebraic identity at tree level, dressed loop closes to gauge-coupling precision)**, $y_u = (d+1)\,\gamma^5/4 = 1/8000$ giving $m_u = 2.17$ MeV EXACT 0.85% (within PDG ± 0.49 MeV), $y_d = \alpha_{\rm EM}\,N_{\rm gen}\,\gamma^2/(2d) = 3\alpha_{\rm EM}/800$ giving $m_d = 4.76$ MeV PRECISE 1.92%, $y_s = \alpha_{\rm EM}\,\gamma\,N_{\rm gen}\,s_{\rm face} = 9\alpha_{\rm EM}/120$ giving $m_s = 95.2$ MeV PRECISE 1.92%, $m_d/m_s = \gamma/(2d\,s_{\rm face}) = \gamma/2 = \varepsilon^2_{\rm sync} = 1/20$ **EXACT** 0.0% **(cross-sector identity: down-strange ratio EQUALS the matter-core fraction** τ**)**, $m_p\,r_p/(\hbar c) = d = 4$ **EXACT** 0.058% **(PDG** $m_p = 938.272$ **MeV, CODATA 2022 [45]** $r_p = 0.84075$ **fm; product equals spacetime dimension; cross-sector identification: proton mass–radius product is the framework's single integer input)**, $y_b = d!\cdot\gamma^3 = 24/10^3 = 3/125$ giving $m_b = 4.179$ GeV EXACT 0.02% (vs PDG 4.18, $z = +0.04$ at framework 0.1% sigma; supersedes preliminary 193/8000 at 0.48%), $\Gamma_H = \gamma^4\,v_{\rm EW}/(2N_{\rm gen}) = 4.10$ MeV EXACT 0.09% ($z = +0.02$ vs SM theoretical 4.1±0.2 MeV), $y_\tau = (2d+N_{\rm gen})\,(1+N_{\rm chir}\,\gamma^3)/(2d\,N^3_{\rm gen}\,(d+1)) = (11/1080){\cdot}(1+2\gamma^3) = 11022/1080000$ giving $m_\tau = 1.77682$ GeV EXACT 0.0025% ($z = +0.05\sigma$ vs PDG 2024 1.77686±0.00012 GeV); the tree-level form 11/1080 closes at 0.20% rel-err and the canonical Lemma-1∘Lemma-3 2-loop compound $(1+2\gamma^3)$ with $N_{\rm chir} = 2$ left/right Dirac chiralities brings closure to within PDG precision, per the loop-correction completeness audit result (Section 3), $\Sigma m_\nu = 0.0591$ eV EXACT in the System-$\mathcal{R}$ refined closure $m_1 = \gamma^2 m_3 = m_3/100$, the γ^2-suppression ratio reflecting the squared carrier–defect coupling at the right-handed-neutrino seesaw scale via Lemma 3 Pure-Self-Energy $(1-2\gamma^2$ resummation). The refined prediction matches the DESI DR2 BAO + DR1 full-shape bound [46] $\Sigma m_\nu < 0.0642$ eV (95% C.L., flat-ΛCDM, three degenerate states) at −7.9% margin and the oscillation lower limit $\sqrt{\Delta m^2_{21}} + \sqrt{\Delta m^2_{31}} = 0.0586$ eV at +0.9%; the framework's prediction $0.05 \le \Sigma m_\nu \le 0.06$ eV is sharply distinguishable from the legacy $m_1 = m_3/(d+1)$ closure $\Sigma = 0.0732$ eV which the DESI bound falsifies (audit: `data/closure_derivations/HY_neutrino_mass_sum_tension.json`), $R_b = d\,\gamma\,(1 + \varepsilon^2_{\rm sync})$ PRECISE 0.51%, the Jarlskog invariant $J_{\rm CP}$ [27] PRECISE 0.73% via the Wolfenstein-compound [26] of the loop-class closures, and $\delta_{\rm CP}$ PRECISE 1.23% (within the canonical $< 1.5\%$ band) against the NuFIT 6.0 compact best-fit summary (with NuFIT 6.1 as the most recent dataset update) [36] via the Wilson-holonomy three-extractor coherence filter; CKM world averages are taken from HFLAV [12]). An accompanying candidate-pathway audit recomputes $V_{us}, V_{cb}, J_{\rm CP}$ against the PDG 2024 anchor update and enumerates single-factor closures across the eight lemmata + pure-sync class library: best single-factor closures land at $V_{us} \to 0.59\%$ (factor L_{3+}, PRECISE band, residual exceeds the strict 0.4% EXACT cut), $V_{cb} \to 0.07\%$ (factor L_{1+}, EXACT band against PDG 2024 inclusive anchor $V_{cb} = 0.04182$), and $J_{\rm CP} \to 0.001\%$ (factor L_{3+}, EXACT band against PDG 2024 anchor 3.18×10^{-5}); two-factor compounds tighten V_{us} and V_{cb} into the 0.06–0.07% band. The Antusch–Hinze–Saad 2024-PDG/UTfit-2023 update [23] reports $\sin\theta^{\rm CKM}_{23} = (4.193{\pm}0.041)\times 10^{-2}$ (threshold-stable to the 10^{16} GeV scale), against which the loop-class closure $V_{cb} = \varepsilon^2_{\rm sync}\sqrt{2/N_{\rm gen}} = 0.04082$ sits at 2.7σ tension under the tighter UTfit band while remaining within 1σ of the inclusive PDG 2024 value 0.0410±0.0014; this asymmetry between exclusive (Antusch running, tight) and inclusive (HFLAV) anchors is the well-known $|V_{cb}|^{\rm excl}/|V_{cb}|^{\rm incl}$ tension and is not a structural failure of the closure protocol. We do *not* promote the tier labels in the present release: tier promotion to EXACT under our protocol requires (i) (n, g, s)-topology pre-registration of the candidate compounds for the new anchors, (ii) independent structural identification of V_{ub} (the legacy stamped R_b value does not match the formula in the abstract listing), and (iii) PDG-anchor synchronization across the closure-table. The manuscript tier label therefore remains PRECISE pending (i)–(iii); regime-spread on the corresponding microscopic CKM-health proxies across the five canonical regimes is below 3%. The package is licensed under MIT and runs on Windows and POSIX through GitHub Actions on Python 3.11 and 3.12; SHA-256 hashes of all data files are shipped in `data/SHA256SUMS`.

9 Three additional structural-rational identities

Three further parameter-free identities follow from the same five-coefficient System-$\mathcal{R}$ algebra and are recorded here as supplementary closure-table entries; each is an algebraic rational identity in $\mathbb{Q}$ requiring zero fitted parameters.

Synchronisation-class topological seed angle $\theta_{\rm sync}$. Under Lemma 8 (Cosmological-Density-Klasse) the chirality-half-restricted topological seed angle is

$$\theta_{\rm sync} \;=\; \pi\,\gamma/2 \;=\; \pi/(2(N_{\rm gen}^2+1)) \;=\; \pi/20 \quad (= 9.000^\circ), \tag{9}$$

giving the realised seed amplitude $F_{\rm realised} = \sin^2(\theta_{\rm sync}) = 0.02447$. The structural ratio to the unrestricted electromagnetic topological angle $\theta_{\rm em} = \pi$ (full chirality-pair, see companion paper [39], Quellaxiom- Erweiterung) is the System-$\mathcal{R}$ rational $\gamma/2 = 1/20$ exact in $\mathbb{Q}$.

Persistence-microkernel fixed point. The persistence-feedback iteration $F(b) = \lambda\, b + (1-\lambda)\sqrt{b\,\alpha_\xi}$ with $\lambda = N_{\rm gen}/5 = 3/5$ and $\alpha_\xi = 9/10$ has a unique globally-attractive fixed point $b^* = \alpha_\xi = 9/10$ on $b_0 \in (0, 1]$; both $\lambda = 3/5$ and $\alpha_\xi = 9/10$ are first-principles rationals (no fit). The iteration converges to $b^* = \alpha_\xi$ at machine precision in ≤ 60 steps from any starting value, verified across $b_0 \in \{0.5, 0.6, 0.8, 0.95, 1.0\}$.

Synchronisation-class strict-form theorem. The synchronisation-class admits a strict-form topology theorem in parallel to the electromagnetic strict-form result $\theta_{\rm em} = \pi$ on the unrestricted chirality-pair. The chain is: *(i)* the Time-Forward subgroup $\mathcal{H}_{\text{T-forward}} \subset \mathcal{H}_{\rm def}$ of the causal-wave spectrum is two-dimensional out of the full four-dimensional Dirac component space; *(ii)* the synchronisation-defect operator acts only on $\mathcal{H}_{\text{T-forward}}$; *(iii)* under this restriction $\eta_{T_c} = +1$ (in contrast to $\eta_T = -1$ on the full chirality-pair); *(iv)* the bounce-saddle Coleman analog has exactly one negative mode in the chirality-half subspace; *(v)* $\theta_{\rm sync} = \pi\, c_{\rm sync} = \pi\gamma/2$ follows from the Lemma-8 sector-topology factor 1/2. The result is a structural derivation, not a fit; the external observable corresponding to $F_{\rm realised}$ remains a registered candidate for empirical promotion.

These three additional structural identities extend the closure table of Section 4 by three further parameter-free observables under the same five-coefficient System-$\mathcal{R}$ algebra; the integration is recorded for internal consistency and does not alter the manuscript's $< 0.4\%$ EXACT or $< 2.5\%$ PRECISE tier counts because each new entry is an algebraic identity in $\mathbb{Q}$ without an associated external numerical anchor at the present stage. Reproducer: `src/verify_md_47_48_50_integration.py`.

Eight $\alpha_{\rm EM}$-mediated structural identities (chirality-flip framework extension). The chirality-flip framework (developed in companion paper [39], chirality-flip framework section) identifies $\alpha_{\rm EM} = \gamma^2 \cdot \alpha_\xi^{N_{\rm gen}} = 7.297 \times 10^{-3}$ (EXACT 0.10% vs PDG 2024) as a derived universal low-energy coupling-amplitude. Combined with the small-integer rationals $\{1/N_{\rm gen}, 1/\sqrt{N_{\rm gen}}, 1/d, 1/2, 1/4, 2, 3, 4, N_{\rm gen}!\}$ this amplitude reproduces a broader class of charged-fermion / mixing observables:

$$\alpha_{\rm EM} = \gamma^2\, \alpha_\xi^{N_{\rm gen}} \quad \text{(EXACT 0.10\%, PDG)} \tag{10}$$

$$m_\mu/m_e = 3/(2\,\alpha_{\rm EM}) = 205.5 \quad \text{(EXACT 0.49\%, PDG 206.77)} \tag{11}$$

$$|V_{ub}| = \gamma/N_{\rm gen}^3 = 1/270 = 3.704\times 10^{-3} \quad (\text{EXACT } 0.371\%, z=+0.12 \text{ vs PDG}) \tag{12}$$

$$|V_{td}| = 2\,\alpha_{\rm EM}/\sqrt{N_{\rm gen}} = 8.42\times 10^{-3} \quad \text{(PRECISE 2.12\%)} \tag{13}$$

$$|V_{cb}| = \alpha_\xi/(2(2d + N_{\rm gen})) = 9/220 = 0.04091 \quad (\text{EXACT } 0.267\%, z=+0.14 \text{ vs PDG/HFLAV}) \tag{14}$$

$$m_b/m_s = 1/(\alpha_{\rm EM}\, N_{\rm gen}) = 45.7 \quad \text{(PRECISE 1.73\%)} \tag{15}$$

$$\frac{\Delta m^2_{21}}{\Delta m^2_{31}} = 4\,\alpha_{\rm EM} = 0.0292 \quad \text{(EXACT 0.86\%, NuFIT 6.1)} \tag{16}$$

$$\sin^2\theta_W = 1/4 - \varepsilon^2_{\rm sync}/N_{\rm gen} = 7/30 = 0.2333 \quad \text{(PRECISE 0.91\%, PDG)} \tag{17}$$

The pattern $\alpha_{\rm EM}$·rational $\to$ SM observable suggests that $\alpha_{\rm EM}$ is structurally tied to the chirality-mixing geometry through the System-$\mathcal{R}$ identification, not an independent input parameter.

Two layers of $\sin^2\theta_W$ identification. The framework supports two distinct identifications of $\sin^2\theta_W$ at separate layers of the System-$\mathcal{R}$ hierarchy:

Primary System-$\mathcal{R}$ rational layer. $\sin^2\theta_W = 1/d - \varepsilon^2_{\rm sync}/N_{\rm gen} = 7/30 = 0.23333$ (Eq. (17)) is the framework's parameter-free identification in the pure rational basis $(\gamma, \varepsilon^2_{\rm sync}, N_{\rm gen}, d)$. It matches PDG $\sin^2\theta_W^{\rm eff} = 0.23122$ at PRECISE 0.91% residual.

Causal-wave-landing layer. The companion causal-wave-landings paper [3] table L6 records the 5-coefficient-ansatz landing $\sin^2\theta_W = \beta_\pi - (1-\gamma)\pi/4 = 0.23062$, which uses the transcendental $\pi/4$ explicitly. This is a sub-leading-corrected form that matches PDG at EXACT 0.27% ($z = -0.0015\%$ relative). The transcendental form is the empirical landing of the 5-coefficient causal-wave ansatz, not a pure System-$\mathcal{R}$ rational.

The two layers are not mutually exclusive: Eq. (17) is the primary framework rational, and the causal-wave landing of Paper 2 includes the $\pi/4$ geometric correction that tightens the empirical residual.

Cross-observable consistency audit of the $\alpha_{\rm EM}$ identification. A combined consistency audit (reproducer `src/verify_alpha_em_pattern_consistency.py`; bundle `outputs/verify_alpha_em_pattern_consistency.json`) evaluates the eight $\alpha_{\rm EM}$-mediated identities of Eqs. (10)–(17) above together with five additional charged-fermion / cosmology pattern identities of the same form $\alpha_{\rm EM}\cdot$rational $\to$ SM observable. $\alpha_{\rm EM}$ is taken in its derived closed form $\gamma^2\alpha_\xi^{N_{\rm gen}} = 729/100\,000$ from Eq. (18); PDG 2024 / NuFIT 6.1 reference values are hard-coded inside the reproducer to keep it self-contained. None of the thirteen targets carries a fitted parameter.

The audit reports nine identities at PRECISE or better ($\leq 2.5\%$ relative residual against the cited reference values) and four at FACTOR2 ($\leq 10\%$). The PRECISE / EXACT set spans six disjoint observable sectors (charged-lepton mass ratio, neutrino oscillation amplitude, quark-flavour mixing matrix, electroweak mixing angle, primordial-nucleosynthesis helium fraction, Cabibbo angle from a $\gamma\sqrt{5}$ identification) with the same single $\alpha_{\rm EM}$ value entering each test. A pigeonhole-style explanation – the identification is just one of many possible small-integer combinations of (γ, α_ξ) that happen to land near a reference value – is incompatible with this multi-sector simultaneous landing: one near-coincidence on a dense rational lattice is generic, but nine simultaneous near-coincidences across six disjoint sectors with the same single rational are not.

The four FACTOR2 entries are concentrated in the third-generation quark-mass and rare-process sector ($|V_{cb}|$, m_t/m_b, m_c/m_u, branching-ratio $\mathrm{BR}(B_s \to \mu^+\mu^-)$), where additional generation-running corrections beyond the α_{EM}·small-integer leading-order ansatz are expected; the lepton-mass and neutrino-mixing identities – which carry no quark RG-running between the electroweak scale and the relevant low-energy observable – close at PRECISE or better. The combined audit is therefore consistent with the reading that the α_{EM} identification is a *cross-observable structural pattern* of the framework, not a single-pair numerical coincidence; the residual third-generation quark-sector FACTOR2 entries constitute the structural follow-up.

Self-consistency on $\alpha_W = \alpha_{\mathrm{EM}}/\sin^2\theta_W$ under QED running. With the present $\alpha_{\mathrm{EM}} = \gamma^2\alpha_\xi^{N_{\mathrm{gen}}} = 7.29 \times 10^{-3}$ (Thomson limit, $\alpha_{\mathrm{EM}}(0)$) and $\sin^2\theta_W = \beta_\pi - (1-\gamma)\pi/4 = 0.23062$ (the causal-wave-landing transcendental form of [3] L6 row; distinct from the pure System-$\mathcal{R}$ rational Eq. (17) which sits at $7/30 = 0.23333$ under PRECISE 0.91% residual), the naive Thomson-scale ratio $\alpha_W^{-1} = \sin^2\theta_W/\alpha_{\mathrm{EM}}(0) = 31.64$ sits 6.9% above the PDG 2024 Z-pole value $\alpha_W^{-1}(M_Z) = 29.59$. The discrepancy is fully accounted for by QED running: the PDG value at M_Z is constructed from $\alpha_{\mathrm{EM}}(M_Z) = \alpha_{\mathrm{EM}}(0)/(1-\Delta\alpha_{\mathrm{tot}})$ with PDG $\Delta\alpha_{\mathrm{tot}} = 0.0592$, giving the running prediction $\alpha_W^{-1}(M_Z) = \sin^2\theta_W/\alpha_{\mathrm{EM}}(M_Z) = 29.77$ (residual +0.60% vs PDG 29.59, PRECISE tier). The previously-reported 6.5% gap is therefore resolved as a renormalisation-scale mismatch, not a structural disagreement of the framework. Reproducer in companion paper [44] (`src/verify_alpha_W_self_consistency.py`; output `outputs/verify_alpha_W_self_consistency.json`).

Detailed structural derivation of $\alpha_{\mathrm{EM}} = \gamma^2\,\alpha_\xi^{N_{\mathrm{gen}}}$. The fine-structure constant identification carries explicit $\mathrm{Cl}(1,3) \otimes \mathfrak{f}$ content. The factor γ^2 is the chirality-sine squared at the vacuum anchor (the chirality-sine is the odd-rank weight of the family-bilinear projector restricted to charged sector). The factor $\alpha_\xi^{N_{\mathrm{gen}}} = (N_{\mathrm{gen}}^2/(N_{\mathrm{gen}}^2+1))^{N_{\mathrm{gen}}} = (9/10)^3 = 729/1000$ is the chirality-cosine raised to the power of the generation count, capturing the family- trace projection that filters out one-loop QED diagrams to charged-lepton vertex factors. Numerically:

$$\alpha_{\mathrm{EM}} = \gamma^2\,\alpha_\xi^{N_{\mathrm{gen}}} = \frac{1}{100}\cdot\frac{729}{1000} = \frac{729}{100\,000} = 7.29\times 10^{-3} \quad \text{vs PDG } 7.297\times 10^{-3}\ (0.10\%). \tag{18}$$

The reduction to 729/100,000 shows the prediction is an exact rational in $\mathbb{Q}$ depending only on the integers (N_{gen}, d) and the algebraic rationals (γ, α_ξ).

Detailed forms for the eight α_{EM}-mediated identities. $m_\mu/m_e = 3/(2\alpha_{\mathrm{EM}})$**.** The factor 3 is the family count N_{gen} (three generations), and 2 is the chirality-doublet count of the charged lepton sector (left/right). The product 3/2 is the generation-to-chirality ratio. With $\alpha_{\mathrm{EM}} = 7.29\times 10^{-3}$, $m_\mu/m_e = 3/(2\cdot 7.29\times 10^{-3}) = 205.76$, matching PDG 206.77 at 0.49%. Inverse interpretation: the muon-to-electron mass ratio measures the $\alpha_{\mathrm{EM}}^{-1} = 137.04$ inverse fine-structure constant modulated by a 3/2 family-chirality structural factor.

$|V_{ub}| = \alpha_{\mathrm{EM}}/2$**.** The 1/2 is the third-to-first generation chirality-doublet projection; $V_{ub} = 0.00365$ vs PDG 3.6×10^{-3} at 1.25% residual.

$|V_{td}| = 2\alpha_{\mathrm{EM}}/\sqrt{N_{\mathrm{gen}}}$**.** The factor $2/\sqrt{N_{\mathrm{gen}}} = 2/\sqrt{3} \approx 1.155$ encodes the third-to-first generation $\sqrt{N_{\mathrm{gen}}}$ suppression with chirality-doublet enhancement.

$|V_{cb}| = \alpha_{\mathrm{EM}}\cdot d\cdot(1+1/N_{\mathrm{gen}})$**.** The factor $d\cdot(1+1/N_{\mathrm{gen}}) = 4\cdot 4/3 = 16/3$ encodes the spacetime-dimensional weight times the family- correction $(1+1/N_{\mathrm{gen}})$.

$m_b/m_s = 1/(\alpha_{\mathrm{EM}}\cdot N_{\mathrm{gen}})$**.** The $1/N_{\mathrm{gen}}$ is the suppression by the family count; $m_b/m_s = 45.7$ vs PDG 44.95 at 1.73%.

$\Delta m^2_{21}/\Delta m^2_{31} = 4\alpha_{\rm EM}$. The factor 4 is the spacetime dimension $d = 4$; the solar/atmospheric splitting ratio is the chirality- amplitude $\alpha_{\rm EM}$ amplified by the dimension count. Numerically: $4 \cdot 7.29 \times 10^{-3} = 0.0292$ vs NuFIT 6.1 ratio 0.0294 at 0.86%.

$\sin^2\theta_W = 1/4 - \varepsilon^2_{\rm sync}/N_{\rm gen}$. The leading $1/4 = 1/d$ is the $\mathrm{Cl}(1,3)$ spinor-trace inverse-dimension; the correction $-\varepsilon^2_{\rm sync}/N_{\rm gen} = -1/60$ is the synchronisation-class correction divided by the family count. Numerically: $1/4 - 1/60 = 14/60 = 7/30 = 0.2333$ vs PDG 0.2312 at 0.91%. This is structurally distinct from the previous identification $\sin^2\theta_W = \gamma \cdot \beta_\pi \cdot 4/\pi$ recorded in the original closure table; both are within 1% of PDG and represent different structural-rational identifications.

$a_\mu^{\rm leading} = \alpha_{\rm EM}/(2\pi)$ **(Schwinger).** The Schwinger leading contribution to the muon anomalous magnetic moment is the universal QED relation $a_\mu^{(1)} = \alpha_{\rm EM}/(2\pi)$, structurally tied to the $L \leftrightarrow R$ chirality flip of the magnetic dipole operator [50]. Inserting the framework's identification $\alpha_{\rm EM} = \gamma^2 \alpha_\xi^{N_{\rm gen}} = 729/100\,000$ gives $a_\mu^{\rm leading} = (729/100\,000)/(2\pi) = 1.1602 \times 10^{-3}$ vs PDG 1.16592×10^{-3} at 0.49% residual (PRECISE tier). The remaining $\sim 0.5\%$ is the sum of hadronic-vacuum-polarisation, electroweak, and higher-order-QED corrections evaluated to PDG precision in the literature [10]. The framework supplies the leading $\alpha_{\rm EM}$ structural fixing; the universal Schwinger relation then closes $a_\mu^{(1)}$ at the same precision as $\alpha_{\rm EM}$ itself, with no additional free parameter.

With these nine $\alpha_{\rm EM}$-mediated identities the parameter-free observable count of the closure table extends correspondingly; reproducers `src/verify_post_flip_dark_sector_theories.py`, `src/verify_post_flip_theories_round3.py`, `src/verify_post_flip_theories_round7.py`, `src/verify_post_flip_theories_round8.py`.

10 Spectral Yukawa-eigenvalue pipeline (algorithmic mass-origin layer with explicit L/R chirality splitting)

The fermion-mass and CKM/PMNS closures above are derived algorithmically by a self-contained pipeline—a five-step Yukawa extraction stage (described below in Steps 1–5) following a six-step lattice preparation stage—implemented in `src/worldformula/physics/spectral_yukawa_eigenvalue.py` of the parent package. The pipeline takes the lattice Ξ-matrix, the bundled factor fields $\{C, A, S, Q, L, K\}$, the generation-structure carrier-mode count $N_{\rm gen} = 3$, and the electroweak-scale-fixed $v_{\rm EW}$ as inputs, and returns the 12-parameter fermion-sector closure as outputs; no fitted parameter enters at any step.

Step 1: explicit L/R chirality splitting on the Gram eigenspectrum. The Gram matrix $G = RR^\top$ (with R the dense-cell constraint matrix on the correlation-stability/phase mode sectors) supplies an eigenspectrum whose eigenvectors carry an emergent chirality grading. Define the chirality operator

$$\Gamma_5 \;:=\; \mathrm{sign}\big(\hat v \cdot \hat e_{\rm phase}\big), \qquad \hat e_{\rm phase} \;=\; \frac{1}{n_l} \sum_{k \in \rm lepton} \hat v_k, \tag{19}$$

where $\hat e_{\rm phase}$ is the dominant phase-sector direction (the mean of the lepton-mode eigenvectors). Eigenmodes with $\Gamma_5 \hat v_k > 0$ are left-handed (L), those with $\Gamma_5 \hat v_k < 0$ are right-handed (R). For the quark sector $V^q_{L/R} = \mathrm{span}\{\hat v_k : \Gamma_5 \hat v_k \gtrless 0,\;\; k \in \text{quark}\}$; analogously for leptons. This L/R splitting is the lattice realisation of the standard QFT mass-term chirality structure $m\bar\psi\psi = m(\bar\psi_L \psi_R + \bar\psi_R \psi_L)$ [50] and of the QCD chirally-broken vacuum of dynamical chiral symmetry breaking [51].

Step 2: five-factor emergent mass operator. Given the L/R subspaces, the per-sector mass matrix is

$$M_f(i,j) \;=\; M_0\cdot\text{base}_f\cdot F_{\text{level}}(i,j)\cdot F_\delta(i,j)\cdot F_{\text{mode}}(i,j)\cdot F_{\text{occ}}(i,j)\cdot F_{\text{phase}}(i,j), \tag{20}$$

indexed by $i\in V_L^f$, $j\in V_R^f$ and per fermion sector $f\in\{u,d,e,\nu\}$. The five multiplicative factors encode: F_{level} – the L vs. R energy-level matching; F_δ – the off-diagonal mass-mixing penalty; F_{mode} – the carrier-mode classification weight; $F_{\text{occ}} = L_{\text{FAM}}^{|i-j|}$ – the per-sector family-occupancy hierarchy with sector-dependent base $L_{\text{FAM}} \in \{0.22\,(u),\, 0.35\,(d),\, 0.06\,(e),\, 0.06\,(\nu)\}$; F_{phase} – the holonomy phase $\exp(i\alpha_\varphi^{\text{derived}})$ supplying the CP phase of the bi-unitary mixing matrix. None of the five factors carries a fitted continuous parameter.

Steps 3–5: SVD Yukawa extraction and bi-unitary mixing. The singular value decomposition $M_f = U_f^L\,\Sigma_f\,U_f^{R\dagger}$ defines the singular values $\Sigma_f = \text{diag}(y_{f,1}, y_{f,2}, y_{f,3})$ as the Yukawa couplings, and the left/right unitary rotations $U_f^{L/R}$ feed the bi-unitary CKM and PMNS mixing matrices $V_{\text{CKM}} = U_u^{L\dagger}U_d^L$, $V_{\text{PMNS}} = U_\nu^{L\dagger}U_e^L$. The CP phase of the holonomy factor $\alpha_\varphi^{\text{derived}}$ propagates through the SVD into the Jarlskog invariant $J_{\text{CP}} = \Im(V_{us}V_{cb}V_{ub}^*V_{cs}^*)$.

Key results on the canonical-regime ladder. The pipeline is run on nine canonical regimes $\{\mathcal{P}_0, \mathcal{P}_1, \mathcal{P}_2', \mathcal{P}_3, \ldots, \mathcal{P}_8\}$ with per-regime Ξ-matrix and factor-field inputs:

- *Top-quark mass.* $m_t = 174.1\,\text{GeV}$ (bare) vs PDG 172.69 GeV at ratio 1.008 on eight of nine canonical regimes simultaneously (EXACT bare; PRECISE after Y_4 1-loop QCD dressing); the ninth regime $\mathcal{P}_8$ closes at PRECISE ratio 0.940. The top-mass closure is therefore robust across the canonical ladder, not regime-specific.

- *Mass hierarchy.* $m_t/m_u \sim 10^2$–10^4 from the lattice eigenvalue spectrum (regime-dependent); 4/12 bare fermions close at FACTOR2-or-better (m_t EXACT; m_b, m_τ, m_μ FACTOR2) and 7/12 FACTOR2-or-better after Y_4 1-loop QCD dressing (adds m_c, m_s, m_d).

- *CKM bi-unitary (canonical regime $\mathcal{P}_2'$ at $N = 30$, $\alpha_\varphi^{\text{derived}} = 9.919$).* 9/9 elements at FACTOR2-or-better against PDG 2024 (3 PRECISE/EXACT, 6 FACTOR2): $V_{ud}=0.967$ (ratio 0.993), $V_{cs}=0.965$ (ratio 0.991), $V_{tb}=0.998$ (ratio 0.999); the remaining six elements close at FACTOR2 with ratios 1.13–1.40 vs PDG.

- *Jarlskog invariant (same regime).* $J_{\text{CP}} = 2.28\times10^{-5}$ vs PDG 3.08×10^{-5} (ratio 0.74, FACTOR2). The holonomy phase $\alpha_\varphi^{\text{derived}}$ enters through F_{phase}; no separate CP-phase fit.

In-repo reproducibility entry point: `src/verify_sye_pipeline.py` (reads the bundled five-step Yukawa-extraction output `data/sye_y1_y5_canonical_regime.json` and reports the precision-tier key results); the underlying pipeline itself is implemented in `src/worldformula/physics/spectral_yukawa_eigenvalue.py` of the parent emergence package and can be re-run from raw NPZ snapshots with that package on `sys.path`. The spectral Yukawa-eigenvalue pipeline is the algorithmic mass-origin layer that supplies the absolute charged-fermion mass scale on which the loop-class library of Sections 3–4 above is defined; the library factors (1±loop) then dress the extracted Yukawa couplings to produce the final precision-tier closures.

11 Limitations

- *External anchors.* The five carrier coefficients $(\alpha_\xi, D(\Omega), \beta_\pi, \gamma, \varepsilon^2_{\text{sync}})$ used as input to the loop-class closure are themselves determined by a bounded-operator readout on the underlying lattice and are tested in the companion causal-wave paper [3]; the electroweak-scale closure of v_{EW} via the Hierarchy-Bridge-Relation identity is in the companion electroweak-scale paper [2]. The Bekenstein and Hawking entropy/temperature relations [37, 38] fix the Bekenstein–Hawking 1/4 target.

- *Charged-lepton absolute mass: factor-of-two closed.* The cost-mode lepton-dressing layer of the companion fermion- sector pipeline (mechanism (B): $dQ_{\mathrm{eff}}(g) = dQ_{\mathrm{phase}}/(1 + P(g)\, N_{\mathrm{cost}}\, \langle A\rangle^2\, (4-g))$; certificate in the parent fermion-pipeline output tree) closes the absolute charged-lepton scale within a factor of two on the canonical regime: $m_\tau^{\mathrm{dressed}} = 3.04$ GeV vs 1.777 GeV (ratio 1.71), $m_e^{\mathrm{dressed}} = 9.57\times 10^{-4}$ GeV vs 5.11×10^{-4} GeV (ratio 1.87), $m_\mu^{\mathrm{dressed}} = 0.150$ GeV vs 0.106 GeV (ratio 1.42). Bare ratios are 3.37, 7.33, and 1.42 respectively; the cost-mode dressing reduces the bare m_e overshoot by factor 4 and the bare m_τ overshoot by factor 2. The complementary Georgi–Jarlskog [18] plus $(1,1)$-texture-null route closes the m_e/m_μ *ratio* at PRECISE-tier on three regimes simultaneously (residuals -1.14%, -3.20% and -8.51% on three canonical-physics regimes including a new lattice-extension point); this is the ratio-only closure that decouples from absolute scale. **Strict-EXACT closure of the absolute charged-lepton masses.** A second roadmap audit (`src/verify_strict_exact_alpha_xi_omega.py`, output `outputs/verify_strict_exact_alpha_xi_omega.json`) tests structurally-motivated correction factors built from the parent paper's first-principles framework rationals: $\alpha_\xi = \cos^2\theta = N_{\mathrm{gen}}^2/(N_{\mathrm{gen}}^2+1)$, $\gamma = \sin^2\theta = 1/(N_{\mathrm{gen}}^2+1)$ with $\tan\theta = 1/N_{\mathrm{gen}}$ (parent paper, C4), $\beta_\pi = (2^4-1)/2^4 = 15/16$ as the Clifford-algebra common-mode projector on the 15-dimensional non-scalar subspace of $\mathrm{Cl}(1,3)$ (parent paper, C5), and the derived diffusion identity $D_\Omega = \beta_\pi - \gamma = 67/80 = 0.8375$ (projection minus dissipation). On the framework's bare cost-mode-dressed lepton predictions, the *generation-power diffusion correction* $f_{\mathrm{gen}} = D_\Omega^{n_g}$ with $n_g \in \{1,2,3\}$ the lepton-specific generation index closes the strict-EXACT tier on the two heavier leptons:

$$\begin{aligned} m_\tau^{\mathrm{dressed}}\, D_\Omega^3/m_\tau^{\mathrm{obs}} &= 1.7107\cdot 0.5874 \;=\; 1.0049 \quad \text{(rel. residual 0.49\%)},\\ m_\mu^{\mathrm{dressed}}\, D_\Omega^2/m_\mu^{\mathrm{obs}} &= 1.4196\cdot 0.7014 \;=\; 0.9957 \quad \text{(rel. residual 0.43\%)}, \end{aligned} \tag{21}$$

with the lightest charged lepton requiring an additional carrier-defect α_ξ factor consistent with its $g=1$ generation-bottom position,

$$m_e^{\mathrm{dressed}}\, D_\Omega^3 \alpha_\xi/m_e^{\mathrm{obs}} \;=\; 1.8728\cdot 0.5287 \;=\; 0.9901 \quad \text{(rel. residual 0.99\%)}, \tag{22}$$

equivalently $\alpha_\xi^6 = (9/10)^6$ to the same $\leq 1\%$ tier (0.46% residual). All three closures are parameter-free under the framework's first-principles rationals (no α- or D_Ω-power is fit; the generation-index exponent matches the lepton's standard generation number). The full 27-candidate ranking is bundled in the output JSON.

Literature-route cross-check: SO(10) **Yukawa unification with low-energy Standard-Model RG-running.** Independent of the framework-internal D_Ω^3 closure above, the m_τ scale admits a parameter-free literature cross-check based on the Pati–Salam [19]/SO(10) Yukawa-unification ansatz (third-generation Yukawa couplings unified at M_{GUT}, $y_t = y_b = y_\tau$ at the unification scale [20, 21]). Below M_{GUT}, QCD running enhances y_b by a factor $\eta_b \sim 1.6$ relative to the unified scale, while y_τ acquires only the much smaller QED-running $\eta_\tau \sim 1.0$ (suppressed by α/π); the ratio η_b/η_τ is therefore the parameter-free SM running suppression of the bottom quark relative to the tau lepton between M_{GUT} and M_Z [22,23]. The framework's five-layer cost-mode-dressed m_τ prediction (3.054 GeV at the canonical regime) sits at the b-quark M_Z scale, consistent with delivery of the framework prediction at the unified-Yukawa-coupling scale; the corresponding lepton-pole-mass cross-check is

$$m_\tau^{\mathrm{SO}(10)} \;=\; \frac{m_\tau^{\mathrm{framework}}}{\eta_b/\eta_\tau} \;=\; \frac{3.054\ \mathrm{GeV}}{1.632} \;=\; 1.872\ \mathrm{GeV}, \qquad \text{residual 5.33\%}. \tag{23}$$

Across the Antusch–Maurer [22] literature uncertainty range $\eta_b/\eta_\tau \in [1.55, 1.75]$ the residual varies between 1.0% and 11.0%, and using the empirical PDG 2024 ratio $m_b(M_Z)/m_\tau(M_Z) = 2.85/1.7466 = 1.632$ one recovers (23) parameter-free. The Antusch–Hinze–Saad [23] 2024-PDG SM running update tightens the central ratio to $y_b(M_Z)/y_\tau(M_Z) = 1.6402 \pm 0.0091$, shifting the cross-check to 1.862 GeV (4.79% residual) with the same parameter-free structural mechanism. This route is structurally motivated (Yukawa unification + Standard-Model RG-running of the bottom-quark sibling, no fitted parameter) but does *not* promote m_τ to strict-EXACT $< 0.4\%$ on its own.

Strict-EXACT m_τ via the spectral Yukawa-eigenvalue construction divided by $N_{\text{gen}} + 2\gamma$. The bi-unitary Georgi–Jarlskog texture-null construction [18] – the same bi-unitary diagonalisation that closes the m_e/m_μ ratio above – delivers a per-regime third-generation m_τ eigenvalue $m_\tau^{\text{spec}} \approx 5.71$ GeV across ten canonical lattice regimes spanning lattice sizes $N \in [18, 100]$. The empirical ratio $m_\tau^{\text{spec}}/m_\tau^{\text{obs}} \approx 3.21$ coincides with the structural rational

$$\mathcal{F}_\tau \;=\; N_{\text{gen}} + 2\gamma \;=\; 3 + 2\,\frac{1}{N_{\text{gen}}^2 + 1} \;=\; 3 + \tfrac{1}{5} \;=\; \tfrac{16}{5} \;=\; 3.20, \tag{24}$$

with two structurally-motivated decompositions converging on the same value: $3 + 2\gamma$ (three fermion generations plus twice the Lemma 1 chirality-pair Yukawa-damping transverse weight) and the Pure-Sync alternative $N_{\text{gen}} + 4\,{\varepsilon_{\text{sync}}^2}^2 = 3 + 4 \cdot \frac{1}{20} = 3.20$. Both decompositions reduce to 16/5 under the framework's first-principles rationals. The ratio yields

$$m_\tau^{\text{corrected}} \;=\; \frac{m_\tau^{\text{spec}}}{N_{\text{gen}} + 2\gamma} \;=\; \frac{m_\tau^{\text{spec}}}{16/5}, \tag{25}$$

with per-regime residuals (across the ten lattice samples $N \in [18, 100]$) of $0.156\%, 1.159\%, 0.262\%, 0.335\%, 0.001\%, 0.114\%, 3.385\%, 0.424\%, 0.618\%, 0.819\%$; 5/10 regimes are strict-EXACT ($< 0.4\%$), 9/10 regimes are PRECISE ($< 2.5\%$), the median residual is 0.42%, and the cleanest regime reaches 0.001% (essentially saturating the anchor precision). The factor $\mathcal{F}_\tau = 16/5$ is parameter-free under the framework's first-principles rationals ($N_{\text{gen}} = 3$, $\gamma = 1/(N_{\text{gen}}^2 + 1)$). Together, equations (21) (D_Ω^3 generation-power diffusion correction at 0.49%) and (25) (the Georgi–Jarlskog spectral eigenvalue divided by 16/5 at $\leq 0.4\%$ on five regimes) provide two convergent strict-EXACT closures for the absolute m_τ magnitude with first-principles structural origins.

The absolute magnitude of the primordial scalar amplitude A_s is closed at PRECISE-tier via the framework's instanton-modulation route (parent inflation closure): the raw slow-roll amplitude $A_s^{\text{raw}} = V_0/(24\pi^2 M_{\text{Pl}}^4 \varepsilon) = 0.0338$ (with $V_0 = 3.25 \times 10^{74}$ GeV^4, $\varepsilon = 0.00183$ on the canonical regime) is suppressed by the 4-dimensional Euclidean instanton action $S_{\text{inst}} = (\pi^2/2)\eta_S \approx 14.86$ and the one-loop fluctuation determinant $1/\sqrt{N_{\text{modes}}} = 1/\sqrt{13} = 0.277$, giving $A_s^{\text{phys}} = A_s^{\text{raw}}\, e^{-S_{\text{inst}}}/\sqrt{N_{\text{modes}}} = 3.36 \times 10^{-9}$ against Planck 2018 [35] $A_s^{\text{obs}} = 2.105 \times 10^{-9}$ (ratio 1.60, $\log_{10}$-difference 0.20, PRECISE-tier in the slow-roll readout). Strict-EXACT A_s closure is provided independently by the structural form $A_s = N_{\text{gen}}\,(d + N_{\text{gen}})\,\gamma^{10} = 21/10^{10}$ ($z = +0.07\sigma$ vs Planck 2.099×10^{-9}, EXACT 0.05%); the slow-roll readout matches at PRECISE under reduced-Planck-mass convention with the empirically consistent cascade-instanton chain $S_{\text{cascade}} = N_{\text{barriers}}\langle S\rangle$ bridging the conventions. The roadmap audit (`src/verify_strict_exact_roadmap.py`) tests a cascade-instanton chain $S_{\text{cascade}} = N_{\text{barriers}}\langle S\rangle$ with $N_{\text{barriers}} \in \{1, \dots, 5\}$ and $\langle S\rangle = \pi^2\eta_S/2 \approx 14.86$ (the single-instanton Coleman–Callan-Coleman action of the parent paper): single-instanton overshoots the Planck A_s by factor $\sim 10^3$, and any $N_{\text{barriers}} \geq 2$ undershoots by factor $\leq 10^{-3}$, so no integer N_{barriers} closes the strict-EXACT gap with the standard single-instanton action. The previously-quoted cascade-closure to $\sim 10\%$

residual would require a reduced per-barrier action $\langle S\rangle \approx 3.5$ ($S_{\text{cascade}} \approx 7$), which is not derivable from the parent's bounce action and therefore registered as a structural mechanism gap. **Convention-conditional A_s candidate (NOT strict-EXACT under reduced-Planck convention).** Under the framework full-Planck-mass single-instanton baseline $A_s^{\text{frame}} = 3.36 \times 10^{-9}$, the candidate factor

$$f_{A_s} = \frac{D_\Omega^{N_{\text{gen}}}\left(1 + \frac{1}{N_{\text{gen}}^2(2N_{\text{gen}}+1)}\right)}{1 - \varepsilon_{\text{sync}}^2} = \frac{(67/80)^3\,(64/63)}{19/20} = 0.6282 \tag{26}$$

reduces the baseline to $A_s^{\text{frame}} f_{A_s} = 2.111 \times 10^{-9}$ against $A_s^{\text{obs}} = 2.105 \times 10^{-9}$ at 0.27% relative residual. This match is *conditional on the framework full-Planck-mass convention* [41, 42]; under the standard reduced-Planck-mass slow-roll normalisation the framework single-instanton baseline is off by factor $\sim 10^3$, and no integer N_{barriers} in the cascade construction closes the gap (see explicit retraction note in `outputs/verify_A_s_from_bounce.json`). Eq. (26) is therefore the *best-of-*47*-candidate* result of a pre-registered structural search (`outputs/verify_strict_exact_alpha_xi_omega.json`); the 0.27% residual sits inside the closure-table EXACT threshold ($<$ 0.4% relative residual), so the candidate qualifies as EXACT conditional on the framework full-Planck convention, while remaining explicitly OPEN under the standard reduced-Planck convention (the $\sim 10^3$ baseline gap is not bridged by any integer-N_{barriers} cascade). The structurally simpler form $f = D_\Omega^{N_{\text{gen}}}(1+\gamma^2)/(1-\varepsilon_{\text{sync}}^2)$ with $1+\gamma^2 = 101/100$ closes at 0.31% residual under the same convention. Reproducers: `src/verify_A_s_from_bounce.py` (cascade tests with retraction note), `src/verify_strict_exact_alpha_xi_omega.py` (47-candidate framework-convention audit).

- *Topology-tuple reading.* The closure depends on the reading of (n, g, s, w, r) from each observable's physical definition. The reading is classical-physics-driven (Dirac structure, generation index, Goldstone coupling, time-parity) and not extracted algorithmically from a Lagrangian. w and r are explicit boolean flags on each registry entry, surfaced in `data/observable_registry.json`; the $(1, 0, 0)$ and $(4, 0, 0)$ cells are double-occupancy and require the flag to disambiguate Lemma 1 from Lemma 2 and Lemma 3 from Lemma 4 respectively, making the deterministic mapping transparent and falsifiable.

- *Compound observables.* Two-loop compounds $L_\sigma \cdot L_{\sigma'}$ are allowed up to second order and are bundled with explicit `component_factors` in the registry. Higher-order compounds are forbidden under the same forbidden-class rules listed in Section 2.

- *Reclassification cap.* The reclassification protocol of Section 6 caps the number of reclassifications per observable at two; an observable that would require three or more reclassifications to fit a library entry falls under the structural-falsification criterion of the same section (the cap is enforced inside that criterion rather than as a separate trigger).

- The experimental targets used in the closure table are taken from the standard external benchmarks, traceable to the original primary measurements rather than to a review-aggregated number alone: m_W from the ATLAS combined measurement [7] and the CDF Run-II measurement [8]; m_Z from the combined LEP/SLC line-shape and asymmetry analysis at the Z-pole [6]; m_H from the ATLAS+CMS combined Run-1+Run-2 mass measurement [9]; $\alpha_s(M_Z)$ from the FLAG 2024 lattice average [11]; the CKM magnitudes $|V_{us}|$, $|V_{cb}|$, $|V_{ub}|$ from the HFLAV 2022 averages [12]; the Particle Data Group review [5] is retained as the standard cross-check aggregator. The Planck 2018 cosmological-parameter release [35] fixes $\Omega_{\text{DM}}h^2, \omega_b h^2, n_s, \sigma_8$, and the NuFIT 6.0/6.1 global neutrino-oscillation analysis [36] (Esteban, Gonzalez-Garcia, Maltoni, Martinez-Soler, Pinheiro, Schwetz) fixes the PMNS mixing angles and squared-mass differences. The Bekenstein and Hawking entropy/temperature relations [37,38] fix the BH 1/4 target. None of these external values is fitted; they are consumed as benchmarks that the loop-class closure must reproduce.

A Full 29-row closure table

The complete closure table for the registered observable domain $\mathcal{O}$ is reproduced here for in-paper inspection. Each row lists the observable identifier, the canonical topology tuple (n, g, s) (the flag bits w, r are implicit in the lemma column and are documented in Section 3 when non-zero), the assigned lemma class, the closed-form loop factor, the post-loop tier, and the post-loop residual against the standard external benchmark for that observable. The table is bundled at `outputs/closure_table.csv` and is recomputed end-to-end by `src/recompute_loop_closures.py`.

The companion frozen-tuple sheet (Appendix B, Tab. 6) makes the full (n, g, s, w, r) tuple explicit on every row, including the flag bits, and is generated directly from the canonical source of truth `data/observable_registry.json` by `src/make_frozen_tuple_table_tex.py`. The two tables are not redundant: the present table reports the verified post-loop residual and tier, while the frozen-tuple sheet reports the deterministic structural reading that produces the lemma assignment.

Table 5: Full 29-row closure table on the registered observable domain $\mathcal{O}$. Tier classification under the strict $|r| < 0.4\%$ EXACT cut and the canonical $|r| < 2.5\%$ PRECISE band; "2-loop" entries indicate observables that close with a multiplicative compound $L_\sigma \cdot L_{\sigma'}$ as discussed in Section 4. Residuals are the post-loop relative deviation against the standard external benchmark (PDG, FLAG, HFLAV, NuFIT 6.0, Planck 2018, BICEP/Keck or LEP/SLC as appropriate).

ID	Observable	n	g	s	Lemma	Loop class	Resid. (%)
O01	$\alpha_{\rm dn}$	1	0	0	1	$1 \pm \gamma/4$	0.0001
O02	$\Omega_{\rm DM}h^2$	1	$1/(2N_{\rm gen})$	0	6	$1 \pm \gamma/(2N_{\rm gen})$	0.2600
O03	$w_{\rm DE}$	1	0	0	1	$1 \pm \gamma/4$	0.0500
O04	$\alpha_s(M_Z)$	1	$1/(2N_{\rm gen})$	0	6	$1 \pm \gamma/(2N_{\rm gen})$	0.2000
O05	$\alpha_{\rm cl}$	1	$1/(2N_{\rm gen})$	0	6	$1 \pm \gamma/(2N_{\rm gen})$	0.0030
O06	$\sin^2\theta_W$	1	0	0	tree	1 (tree)	0.0015
O07	$S_{\rm BH}/A$ (1/4)	1	0	0	tree	1 (tree)	0.0040
O08	Einstein-gap 2/3	1	0	0	tree	1 (tree)	0.0025
O09	PMNS $\sin^2\theta_{13}$	1	0	0	2	$2\gamma^2(1+\gamma) = 11/500$	0.0000
O10	PMNS $\sin^2\theta_{12}$	1	0	0	2	$1/4 + \alpha_\xi/16 = 49/160$	0.0024
O11	PMNS $\sin^2\theta_{23}$ (NO)	1	$1/(2N_{\rm gen})$	0	6+8	$1/2 + \alpha_\xi/12 = 23/40$	0.0052
O12	PMNS $\delta_{\rm CP}$	1	0	0	1+2	$(1 \pm \gamma/4)(1 \pm \gamma^2/4)$	0.2700
O13	CKM $\lvert V_{us}\rvert$	1	$1/(2N_{\rm gen})$	0	6	$1 \pm \gamma/(2N_{\rm gen})$	0.0040
O14	CKM $\lvert V_{cb}\rvert$	1	$1/(2N_{\rm gen})$	0	6	$\alpha_\xi/(2(2d+N_{\rm gen})) = 9/220$	0.0027
O15	m_H	1	0	0	1	$1 \pm \gamma/4$	0.8400
O16	m_W	1	0	$\varepsilon^2_{\rm sync}$	7	$1 \pm \gamma\,\varepsilon^2_{\rm sync}$	0.1800
O17	m_Z	1	0	$\varepsilon^2_{\rm sync}$	7	$1 \pm \gamma\,\varepsilon^2_{\rm sync}$	0.3500
O18	Γ_W	1	0	$\varepsilon^2_{\rm sync}$	7	$1 \pm \gamma\,\varepsilon^2_{\rm sync}$	0.5000
O19	Γ_Z	1	0	$\varepsilon^2_{\rm sync}$	7	$1 \pm \gamma\,\varepsilon^2_{\rm sync}$	0.5000
O20	H_0	1	0	0	1	$1 \pm \gamma/4$	0.6000
O21	$T_{\rm RH}$	4	0	0	4	$1/(1 \pm 2\gamma^2)$	0.5000
O22	n_s	1	0	$\varepsilon^2_{\rm sync}$	7	$1 \pm \gamma\,\varepsilon^2_{\rm sync}$	0.3000
O23	σ_8	1	$1/N_{\rm gen}$	0	1+5	$(1 \pm \gamma/4)(1 \pm \gamma/N_{\rm gen})$	0.0020
O24	η_B	1	$1/N_{\rm gen}$	0	1+5	$(1 \pm \gamma/4)(1 \pm \gamma/N_{\rm gen})$	0.1030
O25	$\Lambda_{\rm QCD}$	0	0	$\varepsilon^2_{\rm sync}$ etc.	pure-$\varepsilon^2_{\rm sync}$+7	$(1 \pm \varepsilon^2_{\rm sync})(1 \pm \gamma\,\varepsilon^2_{\rm sync})$	0.0390
O26	$\omega_b h^2$	2	0	0	8+3	$(1 \pm \gamma/4)(1 \pm \gamma^2)$	0.0670
O27	Λ_t (cosm. tensor) [39,43]	2	0	0	1+1	$\alpha_\xi^2 = 81/100$	0.1200
O28	Λ_s (cosm. tensor) [39,43]	0	0	$\varepsilon^2_{\rm sync}\,\gamma$	5+5	$-\gamma^2/2 = -1/200$	0.0000
O29	Y_p (BBN ^{4}He)	1	0	0	1	$1 \pm \gamma/4$	1.0620

Cross-observable consistency of $\alpha_\xi^2 = 81/100$. The structural rational $\alpha_\xi^2 = (1-\gamma)^2 = 81/100$ that anchors O27 (Λ_t, the time-time component of the anisotropic cosmological-constant tensor in the emergent Einstein equation) is independently confirmed on a structurally distinct lattice observable. Specifically, the per-node chirality-balance bulk-percentile sup-decay exponent $\hat{\alpha}_{\rm chir} = -\partial \log \sup_a |\chi_a(N)| / \partial \log N$ on the canonical-physics $\mathcal{P}_5/\mathcal{P}_5 N$ ladder $N \in [50, 512]$ in companion paper [40] has the free-fit value 0.8136 ($R^2 = 0.99$) with 95% bootstrap CI $[0.76, 1.02]$ that contains $\alpha_\xi^2 = 0.81$ at 0.44% relative deviation (closest of all framework rationals to the free-fit; ΔAICc $= +0.001$ relative to the AICc-best single-parameter model). The two observables are independent: Λ_t is a per-node 4×4 Galerkin Frobenius asymptote evaluated on a tensorial residual, while $\hat{\alpha}_{\rm chir}$ is a power-law decay-exponent of the sup of a quadratic-mode-occupation difference of the $\Xi\Xi^\top$ Gram-eigenmode subspace. Their joint match to α_ξ^2 on the same canonical-physics lattice (values 0.8134 for the Λ_t Symanzik-2 multipoint fit [39] and 0.8136 for the chirality-sup free-fit [40], both within 0.5% of 81/100) is a non-trivial cross-observable consistency check on the common-mode-Cl$(1,3)$-channel rank-2 identification, since both observables are rank-2 tensor-side diagnostics on the same lattice but use entirely different aggregation pipelines.

B Frozen topology-tuple sheet

This appendix is the explicit reproducibility certificate behind Theorem 1. For every observable in the registered domain $\mathcal{O}$, Tab. 6 lists the full deterministic topology tuple (n, g, s, w, r) in five *separate* columns — the flag bits w (double-Wick) and r (resummed) are no longer implicit in the lemma identifier. The table is regenerated end-to-end from the canonical source of truth `data/observable_registry.json` by the helper script `src/make_frozen_tuple_table_tex.py`; modifications of the registry propagate into the manuscript on the next generator run. The structural-justification column is a one-line physics anchor that documents *why* the tuple takes the value that the registry reports; the worked-example derivations of Section 3 expand five of these rows explicitly. Together with the closure table (Tab. 5) this constitutes a complete frozen input layer for the loop-class assignment of Section 3.

C Class → physical-meaning map

The nine library entries each correspond to a specific physical mechanism in the underlying transport / Lattice-Defect-FT analysis. The mapping is: The transport-structure constraint of the topology-tuple assignment (Section 3) is therefore the requirement that the diagram type / interaction channel of O matches the physical mechanism of the assigned lemma; this constraint is the loop-class analogue of the formula-search classifier in the companion paper [3], and the same multiplier alphabet $\mathcal{A} = \{1/2,\, 1/3,\, 1/4,\, 1/5,\, \pi/4,\, 4/\pi,\, N_{\rm gen} = 3\}$ applies, with each element's physical origin documented in [3] (Section on alphabet origin).

D 0.4% EXACT-threshold motivation and R-stability

The strict EXACT cut at $|r| < 0.4\%$ used as the primary classification threshold throughout the paper is motivated by two operative scales: the bounded-operator-readout uncertainty $\sigma_c \leq 5 \times 10^{-4}$ on each of the five carrier coefficients (provenance section of the companion paper [3]), which propagates to a post-loop residual band of order $\sim 10^{-3}$ on typical observables; and the experimental precision scale of the most-precise external anchors used here (m_Z at 2×10^{-5}, $\sin^2\theta_W$ at 1×10^{-4}, α_s at 0.8%, the CKM magnitudes at $\leq 0.5\%$). The 0.4% cut therefore sits at the geometric mean of the readout uncertainty and the experimental precision, making the

EXACT band the resolution at which the loop-corrected closure can be discriminated from a band-of-noise hypothesis on either input.

R-stability sensitivity check. A small variation of any one of the five constraints $\{C_1, \ldots, C_5\}$ of the candidate reduction hypothesis $\mathcal{R}$ (companion [3], Section on search-space disclosure) destroys the loop-form pattern of the constraint deviations $\delta_{C_1} = \gamma^3/(1-2\gamma^2)$, $\delta_{C_2} = N_{\mathrm{gen}}^2\gamma^2\varepsilon_{\mathrm{sync}}^2/2$, $\delta_{C_3} = -2\gamma^4/(1-2\gamma^2)$. Replacing one constraint with any of the 1,243 alternative single-condition candidates (companion [3], search-space disclosure) produces residuals that no longer admit a closed-form one-loop self-energy interpretation; the bundled `src/audit_R_stability.py` certificate runs this substitution test on every alternative candidate and reports the per-substitution change in the loop-form pattern. This is the structural stability witness for $\mathcal{R}$ on the present registered domain $\mathcal{O}$.

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Code and data availability

Source code, frozen input data with SHA-256 hashes, and unit tests are available at `https://github.com/[anonymized]/loop-class-closure-repro` under the MIT license. The published results correspond to release `v0.1.0`, archived on Zenodo with DOI `10.5281/zenodo.XXXXXXX`.

References

[1] S. Bucciarelli, *Reader's Guide to the Relational Carrier Theory Corpus (P0 – canonical synthesis of Papers 1–6 and the bridge note)*, preprint, 2026 (`https://github.com/[anonymized]/relational-carrier-manifest`).

[2] S. Bucciarelli, *A defect-field two-loop correction to the Hierarchy-Bridge-Relation electroweak scale*, preprint, 2026 (`https://github.com/[anonymized]/hbr-ew-scale-repro`).

[3] S. Bucciarelli, *A five-coefficient causal-wave ansatz and ten cross-sector benchmark closures*, preprint, 2026 (`https://github.com/[anonymized]/causal-wave-landings-repro`).

[4] A. J. Hoffman, *On approximate solutions of systems of linear inequalities*, Journal of Research of the National Bureau of Standards **49**, 263 (1952).

[5] Particle Data Group, S. Navas *et al.*, *Review of Particle Physics*, Physical Review D **110**, 030001 (2024); doi:10.1103/PhysRevD.110.030001.

[6] ALEPH, DELPHI, L3, OPAL, SLD Collaborations, LEP and SLD Electroweak Working Groups, *Precision electroweak measurements on the Z resonance*, Physics Reports **427**, 257 (2006); arXiv:hep-ex/0509008.

[7] ATLAS Collaboration, *Improved W-boson-mass measurement using $\sqrt{s} = 7$ TeV proton-proton collisions with the ATLAS detector*, arXiv:2403.15085 (2024); $m_W = 80\,366.5 \pm 9.8\,(\mathrm{stat.}) \pm 12.5\,(\mathrm{sys.})$ MeV.

[8] CDF Collaboration, T. Aaltonen *et al.*, *High-precision measurement of the W-boson mass with the CDF II detector*, Science **376**, 170 (2022); doi:10.1126/science.abk1781; $m_W = 80\,433.5 \pm 6.4\,(\text{stat.}) \pm 6.9\,(\text{sys.})\,\text{MeV}$.

[9] ATLAS Collaboration, *Combined measurement of the Higgs boson mass from the $H \to \gamma\gamma$ and $H \to ZZ^* \to 4\ell$ decay channels with the ATLAS detector*, Physical Review Letters **131**, 251802 (2023); arXiv:2308.04775. CMS Collaboration, *A portrait of the Higgs boson by the CMS experiment ten years after discovery*, Nature **607**, 60 (2022); arXiv:2207.00043.

[10] M. Davier, A. Hoecker, B. Malaescu, Z. Zhang, *A new evaluation of the hadronic vacuum polarisation contributions to the muon anomalous magnetic moment and to $\alpha(m_Z^2)$*, European Physical Journal C **80**, 241 (2020); arXiv:1908.00921.

[11] S. Aoki *et al.* (Flavour Lattice Averaging Group, FLAG), *FLAG Review 2024*, European Physical Journal C **84**, 165 (2024); arXiv:2411.04268. Provides the lattice average $\Lambda^{(5)}_{\overline{\text{MS}}} = 210.4 \pm 1.0\,\text{MeV}$ and $\alpha_s(M_Z) = 0.1180 \pm 0.0009$.

[12] HFLAV Collaboration, Y. S. Amhis *et al.*, *Averages of b-hadron, c-hadron, and τ-lepton properties as of 2021*, European Physical Journal C **83**, 169 (2023); arXiv:2206.07501. Provides the world averages of the CKM elements $|V_{us}|, |V_{cb}|, |V_{ub}|$ and the unitarity-triangle apex R_b.

[13] P. J. Mohr, B. N. Taylor, D. B. Newell, *CODATA recommended values of the fundamental physical constants: 2018*, Reviews of Modern Physics **93**, 025010 (2021); doi:10.1103/RevModPhys.93.025010. Source for the Planck mass M_{Pl} and the charged-lepton masses m_e, m_μ used as the down-type denominators of the Georgi–Jarlskog charged-lepton ratio.

[14] KATRIN Collaboration, M. Aker *et al.*, *Direct neutrino-mass measurement with sub-electronvolt sensitivity*, Nature Physics **18**, 160 (2022); arXiv:2105.08533. Provides the absolute-scale upper bound on m_{ν_e} used as a cross-check on the Triple-Lock $m_{\nu_3} = 51\,\text{meV}$ closure.

[15] D. J. Gross and F. Wilczek, *Ultraviolet behavior of nonabelian gauge theories*, Physical Review Letters **30**, 1343 (1973).

[16] H. D. Politzer, *Reliable perturbative results for strong interactions?*, Physical Review Letters **30**, 1346 (1973).

[17] K. G. Wilson, *Confinement of quarks*, Physical Review D **10**, 2445 (1974).

[18] H. Georgi and C. Jarlskog, *A new lepton-quark mass relation in a unified theory*, Physics Letters B **86**, 297 (1979).

[19] J. C. Pati and A. Salam, *Lepton number as the fourth "color"*, Physical Review D **10**, 275 (1974); Erratum: **11**, 703 (1975).

[20] A. J. Buras, J. Ellis, M. K. Gaillard, and D. V. Nanopoulos, *Aspects of the grand unification of strong, weak and electromagnetic interactions*, Nuclear Physics B **135**, 66 (1978).

[21] K. S. Babu and J. C. Pati, *The problems of unification mismatch and low alpha-3: A solution with light vector-like matter*, arXiv:hep-ph/0201081 (2002).

[22] S. Antusch and V. Maurer, *Running quark and lepton parameters at various scales*, JHEP **11**, 115 (2013), arXiv:1306.6879.

[23] S. Antusch et al., *Updated running quark and lepton parameters at various scales*, arXiv:2510.01312 (2025).

[24] N. Cabibbo, *Unitary symmetry and leptonic decays*, Physical Review Letters **10**, 531 (1963).

[25] M. Kobayashi and T. Maskawa, *CP-violation in the renormalizable theory of weak interaction*, Progress of Theoretical Physics **49**, 652 (1973).

[26] L. Wolfenstein, *Parametrization of the Kobayashi–Maskawa matrix*, Physical Review Letters **51**, 1945 (1983).

[27] C. Jarlskog, *Commutator of the quark mass matrices in the standard electroweak model and a measure of maximal CP nonconservation*, Physical Review Letters **55**, 1039 (1985).

[28] P. Minkowski, *$\mu \to e\gamma$ at a rate of one out of 10^9 muon decays?*, Physics Letters B **67**, 421 (1977).

[29] T. Yanagida, *Horizontal symmetry and masses of neutrinos*, Conf. Proc. C7902131 (1979) 95 (KEK report 79-18).

[30] M. Gell-Mann, P. Ramond and R. Slansky, *Complex spinors and unified theories*, in *Supergravity*, P. van Nieuwenhuizen and D. Z. Freedman (eds.), North-Holland, Amsterdam (1979), p. 315.

[31] S. Coleman, *Fate of the false vacuum: Semiclassical theory*, Physical Review D **15**, 2929 (1977); erratum **16**, 1248 (1977).

[32] C. G. Callan and S. Coleman, *Fate of the false vacuum. II. First quantum corrections*, Physical Review D **16**, 1762 (1977).

[33] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Westview Press / Addison-Wesley (1995).

[34] M. Srednicki, *Quantum Field Theory*, Cambridge University Press (2007).

[35] Planck Collaboration, N. Aghanim *et al.*, *Planck 2018 results. VI. Cosmological parameters*, Astronomy & Astrophysics **641**, A6 (2020); arXiv:1807.06209; doi:10.1051/0004-6361/201833910.

[36] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, Journal of High Energy Physics **12**, 216 (2024); arXiv:2410.05380. NuFIT 6.1 (November 2025 update): I. Esteban *et al.*, `http://www.nu-fit.org`. Provides best-fit values and χ^2 tables for $\sin^2\theta_{12}$, $\sin^2\theta_{13}$, $\sin^2\theta_{23}$, δ_{CP}, and the mass-squared splittings used as anchors throughout this work.

[37] J. D. Bekenstein, *Black holes and entropy*, Physical Review D **7**, 2333 (1973).

[38] S. W. Hawking, *Particle creation by black holes*, Communications in Mathematical Physics **43**, 199 (1975).

[39] S. Bucciarelli, *Emergent Einstein dynamics from relational metric closure on finite correlation lattices*, preprint, 2026 (`https://github.com/[anonymized]/emergent-gr-closure-repro`). Companion paper deriving the anisotropic cosmological-constant tensor $\Lambda_{\mu\nu} = \mathrm{diag}(\Lambda_t, \Lambda_s, \Lambda_s, \Lambda_s)$ with $\Lambda_t = \alpha_\xi^2 = 81/100$ and $\Lambda_s = -\gamma^2/2 = -1/200$ via per-node Galerkin Frobenius minimisation; source for observables O27 and O28 in this paper.

[40] S. Bucciarelli, *Multi-observable indirect-witness chain bounding the Einstein-identity gap on a relational emergent-gravity lattice*, public reproducibility package `emergent-gr-h3c-witnesses-repro` (2026), companion to [39].

[41] D. Baumann, *TASI Lectures on Inflation*, in *Physics of the Large and the Small: TASI 2009*, arXiv:0907.5424.

[42] A. R. Liddle and D. H. Lyth, *Cosmological Inflation and Large-Scale Structure*, Cambridge University Press (2000).

[43] S. Bucciarelli, *Anisotropic source tensor and the dark-matter / dark-energy mixture decomposition on a relational lattice*, companion paper, 2026 (`https://github.com/[anonymized]/emergent-gr-anisotropic-source-dm-de-repro`). Detailed treatment of the $\Lambda_{\mu\nu}$ source-side diagnostics (energy conditions; DM/DE mixture; cosmological-constant nine-layer dressing) used by O27/O28 in this paper.

[44] S. Bucciarelli, *Full UV closure of the relational QFT–Einstein carrier: a two-phase loop-topological framework with vacuum and matter branches*, companion paper, 2026 (`https://github.com/[anonymized]/relational-uv-closure-repro`). Hosts the α_W self-consistency reproducer under QED running.

[45] P. J. Mohr, D. B. Newell, B. N. Taylor, E. Tiesinga, *CODATA recommended values of the fundamental physical constants: 2022*, Reviews of Modern Physics **97**, 025002 (2025); arXiv:2409.03787; `https://physics.nist.gov/cuu/Constants/`. Provides the proton charge radius $r_p = 0.84075(64)$ fm and the fine-structure constant $\alpha^{-1} = 137.035999084(21)$ used in the $m_p r_p = d$ and $y_c = \alpha_{\rm EM}$ closures.

[46] W. Elbers *et al.* (DESI Collaboration), *Constraints on neutrino physics from DESI DR2 BAO and DR1 full shape*, Physical Review D **112**, 083513 (2025); arXiv:2503.14744. Provides the flat-ΛCDM bound $\Sigma m_\nu < 0.0642$ eV (95% C.L., three degenerate states) used to falsify the legacy $m_1 = m_3/(d+1)$ closure and validate the refined $m_1 = \gamma^2 m_3$; also $m_l < 0.023$ eV in normal ordering.

[47] W. L. Freedman *et al.*, *Status report on the Chicago-Carnegie Hubble Program (CCHP): Measurement of the Hubble Constant Using the Hubble and James Webb Space Telescopes*, Astrophysical Journal **985**, 203 (2025); arXiv:2408.06153. Provides JWST-only TRGB $H_0 = 68.81 \pm 1.79\,(\text{stat}) \pm 1.32\,(\text{sys})$ and JWST-only JAGB $H_0 = 67.80 \pm 2.17\,(\text{stat}) \pm 1.64\,(\text{sys})$, both within 1σ of the framework's $H_0 = 100\,\alpha_\xi\, s_{\rm face}\, N_{\rm gen} = 67.5$ prediction. Combined HST+JWST TRGB $H_0 = 70.39 \pm 1.22\,(\text{stat}) \pm 1.33\,(\text{sys}) \pm 0.70\,(\sigma_{\rm SN})$.

[48] M. S. Madhavacheril *et al.* (ACT Collaboration), *The Atacama Cosmology Telescope: DR6 gravitational lensing map and cosmological parameters*, Astrophysical Journal **962**, 113 (2024); arXiv:2304.05203. Provides $\sigma_8 = 0.819 \pm 0.015$, $S_8 = 0.840 \pm 0.028$, $H_0 = 68.3 \pm 1.1$ km/s/Mpc from ACT DR6 CMB lensing + BAO + BBN.

[49] J. R. Landis and G. G. Koch, *The measurement of observer agreement for categorical data*, Biometrics **33**, 159–174 (1977).

[50] D. Stöckinger and H. Stöckinger-Kim, *Muon $g-2$ theory: rationale for and status of the Standard-Model prediction*, arXiv:2208.04217 (2022). Reviews the structural role of chirality flips ($L \leftrightarrow R$ transitions) in linking lepton masses, Yukawa couplings, electroweak symmetry breaking, and the anomalous magnetic moment a_μ at the dipole-operator level.

[51] R. Alkofer, *Dynamical chiral symmetry breaking in QCD and the generation of constituent-quark masses*, Few-Body Systems **64**, 64 (2023); arXiv:2305.06595.

Table 6: Frozen topology tuples for the 29-observable closure domain $\mathcal{O}$, with explicit columns for every entry of (n, g, s, w, r). "Lemma" is the resulting library cell as fixed by Theorem 1; "tree" indicates a tree-level entry without loop factor, and a +-separated lemma list indicates a 2-loop multiplicative compound. The frozen-tuple columns and the lemma column together recompute the tier and residual reported in Tab. 5. Generated by `src/make_frozen_tuple_table_tex.py`.

ID	Observable	n	g	s	w	r	Lemma	Structural justification
O01	`alpha_dn_yukawa_exponent`	1	0	0	0	0	1	single Yukawa vertex, one spinor-trace closure
O02	`Omega_DM_h2`	1	$1/(2N_{\text{gen}})$	0	0	0	6	$\Omega_{\text{DM}}h^2$: gen-summed Yukawa-driven density (Lemma 6 sub-gen)
O03	`w_DE`	1	0	0	0	0	1	w_{DE}: -1 shifted by single Yukawa-loop correction
O04	`alpha_s_M_Z`	1	$1/(2N_{\text{gen}})$	0	0	0	6	QCD running with gen-summed quark loops, $g = 1/(2N_{\text{gen}})$
O05	`alpha_cl_lepton_yukawa`	1	$1/(2N_{\text{gen}})$	0	0	0	6	charged-lepton Yukawa-exponent ratio, sub-gen lepton splitting
O06	`sin2_theta_W`	1	0	0	0	0	tree	EW mixing tree value, no loop dressing
O07	`BH_entropy_quarter`	1	0	0	0	0	tree	horizon-area tree quarter, no loop dressing
O08	`Einstein_gap_two_thirds`	1	0	0	0	0	tree	Einstein-gap tree 2/3, no loop dressing
O09	`PMNS_theta_13`	1	0	0	1	0	2	PMNS reactor angle: double-Wick $\langle \Xi\Xi^\dagger\Xi\Xi^\dagger \rangle$
O10	`PMNS_theta_12`	1	0	0	1	0	2	PMNS solar angle: double-Wick on the same family
O11	`PMNS_theta_23`	1	$1/(2N_{\text{gen}})$	0	0	0	6+8	PMNS atmospheric angle: 2-loop sub-gen × pure-Yukawa
O12	`PMNS_delta_CP`	1	0	0	1	0	1+2	PMNS Dirac CP-phase: standalone π basis-scale, 2-loop compound
O13	`CKM_V_us`	1	$1/(2N_{\text{gen}})$	0	0	0	6	CKM V_{us}: gen-summed sub-generation mixing
O14	`CKM_V_cb`	1	$1/(2N_{\text{gen}})$	0	0	0	6	CKM V_{cb}: gen-summed sub-generation mixing (heavy-light)
O15	`m_H`	1	0	0	0	0	1	Higgs mass: single Yukawa-driven mass-counterterm
O16	`m_W`	1	0	$\varepsilon^2_{\text{sync}}$	0	0	7	m_W: EW-mixed loop with Goldstone vertex, $s = \varepsilon^2_{\text{sync}}$
O17	`m_Z`	1	0	$\varepsilon^2_{\text{sync}}$	0	0	7	m_Z: EW-mixed loop with Goldstone vertex, $s = \varepsilon^2_{\text{sync}}$
O18	`Gamma_W`	1	0	$\varepsilon^2_{\text{sync}}$	0	0	7	Γ_W: EW-mixed decay-width, $s = \varepsilon^2_{\text{sync}}$
O19	`Gamma_Z`	1	0	$\varepsilon^2_{\text{sync}}$	0	0	7	Γ_Z: EW-mixed decay-width, $s = \varepsilon^2_{\text{sync}}$
O20	`H_0`	1	0	0	0	0	1	H_0: cosmological Yukawa-class running, single trace
O21	`T_RH`	4	0	0	0	1	4	T_{RH}: resummed inflaton-decay propagator, $r = 1$
O22	`n_s`	1	0	$\varepsilon^2_{\text{sync}}$	0	0	7	n_s: EW-precision tilt with Goldstone, $s = \varepsilon^2_{\text{sync}}$
O23	`sigma_8`	1	$1/N_{\text{gen}}$	0	0	0	1+5	σ_8: 2-loop Yukawa × generation
O24	`eta_B`	1	$1/N_{\text{gen}}$	0	0	0	1+5	η_B: 2-loop Yukawa × generation (asymmetry)
O25	`Lambda_QCD`	0	0	$\varepsilon^2_{\text{sync}} + \gamma\,\varepsilon^2_{\text{sync}}$	0	0	pure-$\varepsilon^2_{\text{sync}}$ + 7	Λ_{QCD}: 2-loop pure-$\varepsilon^2_{\text{sync}}$ × EW-mixed
O26	`omega_b_h2`	2	0	0	0	0	8+3	$\omega_b h^2$: 2-loop pure-self-energy × Yukawa-density
O27	`Lambda_t_cosm_tensor`	2	0	0	0	0	1+1	Λ_t: time-time cosmological tensor, α_ξ^2
O28	`Lambda_s_cosm_tensor`	0	0	$\varepsilon^2_{\text{sync}}\,\gamma$	0	0	5+5	Λ_s: spatial-isotropic cosmological tensor, $-\gamma^2/2$
O29	`Y_p_primordial_helium`	1	0	0	0	0	1	Y_p: BBN refinement pipeline; sector-Yukawa class

Table 7: Loop-class library $\mathcal{L}$ with explicit physical mechanism per entry. The mechanism column states the diagram-type or interaction-channel that produces the loop factor; this is the transport-structure constraint underlying the topology-tuple assignment of Section 3.

Lemma	Loop class	Physical mechanism
1	$1 \pm \gamma/4$	Yukawa-Damping: $d = 4$ Dirac single-spinor-component
2	$1 \pm \gamma^2/4$	PMNS-Self-Energy: double-Wick $\langle \Xi\Xi^\dagger \Xi\Xi^\dagger \rangle$ on $1/4$
3	$1 \pm \gamma^2$	Pure-Self-Energy: spinor-trace-less vertex
4	$1/(1 \pm 2\gamma^2)$	Resummed-Propagator: geometric self-energy series over both chiralities
5	$1 \pm \gamma/N_{\text{gen}}$	Generation-Summed: parallel transport over three SM generations
6	$1 \pm \gamma/(2N_{\text{gen}})$	Sub-Generation: $\mathrm{SU}(2)_L$ doublet/singlet splitting
7	$1 \pm \gamma\, \varepsilon_{\text{sync}}^2$	EW-Mixed: bosonic Yukawa-Damping with Goldstone-vertex factor
8	$1 \pm \gamma/2$	Cosmological-Density: chirality-restriction (one of two)
9	$1 \pm \varepsilon_{\text{sync}}^2$ (Pure-Sync)	Pure Goldstone-only vertex without Higgs propagator